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A Survey on Loss-based Regularization for Model Generalization

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Abstract
Regularization is a central and fundamental research topic in machine learning. Despite the numerous methods developed in current fast growing literature, there is a lack of theoretical understanding and rigorous analysis of regularization. The first part of this paper is an answer to this need: we formulate regularization as a solution of ill-posed problems, examine fundamental principles of regularization, analyze the effectiveness of Tikhonov regularization in linear systems, and propose a consistency principle for the design of regularization methods in both linear and nonlinear systems. Based on these analyses, the second part of this paper is a survey of recent loss-based regularizers. We develop a comprehensive taxonomy and summarize prominent features of these regularizers. Finally, results of benchmark tests confirm the effectiveness of representative loss-based regularizers.

Keywords Loss-based regularization · Model generalization · Empirical risk minimization · Inner-product spaces · Singular value decomposition · Tikhonov regularization · Consistency principle

1 Introduction

Regularization is a ubiquitous process for solving ill-posed problems in mathematics, statistics, inverse problems, and machine learning (ML). In ML, regularization is an effective way to prevent overfitting (the training data) and to improve model generalization (on the unseen data).

In recent years, regularization and model generalization have been intensively studied (Zhang et al., 2021; Keskar et al., 2019; Jacot et al., 2018; Arora et al., 2019; Belkin et al., 2019; Yang et al., 2020). Many methods are based on the heuristic principle that simple models with flat minimums are less prone to overfit, e.g., Hochreiter and Schmidhuber

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(Hochreiter and Schmidhuber, 1997) reported that a parametric model with a flat local minimum has better generalization. Another approach is the data augmentation that expands the capacity of a dataset by generating dummy data and improves model generalization via reducing its dependence on specific attributes. For image data, it increases the number of training samples by randomly translating, cropping, flipping, etc. Recent techniques of data augmentation include the mixup (Zhang et al., 2018) and the noisy feature mixup (Lim et al., 2022). There exist many other regularization techniques such as batch normalization (BN) (Dauphin and Cubuk, 2021; Ergen et al., 2022), adversarial training (Chan et al., 2020), and the early stop strategy (Lodwich et al., 2009). Weight decay (Lang and Hinton, 1989) alleviates overfitting by penalizing the L_2 norm of weight parameters. Dropout (Srivastava et al., 2014; Wei et al., 2020) reduces the size of a neural network by randomly dropping partial neurons with a certain probability during each training procedure.

As a topic of fundamental importance and high popularity, regularization methods have been frequently reviewed in recent years (Shorten and Khoshgoftaar, 2019; Labach et al., 2019; Hu et al., 2021; Kukačka et al., 2018; Moradi et al., 2020; Santos and Papa, 2022; Tian and Zhang, 2022). Shorten and Khoshgoftaar (2019) and Labach et al. (2019) surveyed data augmentation and dropout in deep learning, respectively. Hu et al. (2021) concentrated on low-rank regularization and its applications. Kukačka et al. (2018) suggested a framework for categorizing current regularization strategies in deep learning. Moradi et al. (2020) reviewed regularization in deep learning. Santos and Papa (2022) focused on regularization of convolutional neural networks. Tian and Zhang (2022) discussed ML regularizers, their distinguishing features, application scenarios, and solution algorithms.

An open problem of regularization is how to theoretically analyze their effectiveness. Even how regularization relates to traditional mathematics (such as approximation theory and functional analysis) has rarely been addressed in a self-contained and coherent manner. This is a key motivation of this paper, where we approach ML regularization from a theoretical viewpoint, explain key regularization concepts with traditional mathematics, review recent loss-based regularizers, and summarize their practical aspects.

The main contributions of this paper are as follows.

A. The concept of regularization is formulated as a solution to ill-posed problems on the solid ground of inner-product spaces. Two fundamental principles of regularization, viz. the spectral cutoff and the Tikhonov regularization, are rigorously examined and explained with well-known techniques in statistics, numerical analysis, and linear algebra.
B. The effectiveness of Tikhonov regularization for linear models is rigorously analyzed: we recall the classical equivalence with L_2-penalized least squares and prove MSE reduction under suitable conditions; we also propose a consistency principle as a guideline for designing regularizers. A and B form a self-contained coherent exposition on theoretical aspects of regularization.
C. We survey recent loss-based regularizers and categorize them according to the penalized components of the learning process. We summarize the prominent features and application scenarios of these regularizers in several tables for a clear overview and easy lookup.

This article focuses on *loss-based* regularization, i.e., techniques that alter the empirical risk or add explicit penalties to the loss. We do not cover architecture- or training-procedure regularizers (e.g., dropout, batch normalization, data augmentation), which act primarily through *implicit* regularization and optimization dynamics. Our rationale is twofold: (i) loss-based methods admit a unified analysis via stability/consistency and degrees-of-freedom arguments; and (ii) they provide portable and task-agnostic knobs that can be turned without architectural changes. We briefly relate loss-based and implicit/architecture regularization at the end of Section 4 (before Figure 2).

The rest of this paper is organized as follows. In Section 2, we approach ML via empirical risk minimization. In Section 3, we precisely define and fully explain key concepts of regularization to lay down the ground for rigorously analyzing the effectiveness of explicit regularizers and for proposing a consistency principle. In Section 4, we discuss several key concepts and heuristics for designing nonlinear regularizers, painting a roadmap for the rest of this paper. In the next three sections, we review recent loss-based regularization methods according to the component of the learning process that is being penalized: the weights (Section 5), the output/input gradient (Section 6), and the empirical risk loss (ERL) function (Section 7). We summarize main features and application scenarios of these regularizers in several tables in Section 8 and conclude this paper in Section 9.

2 Empirical Risk Minimization (ERM)

An ML system consists of three major components: data, models, and learning. Data represent a real-world process that calls for a simple model while learning means finding the parameters of the model, hence it is the models that lie at the core of ML. In this work, we adopt the mathematical viewpoint that an ML model is a predictive function (or simply a *predictor*) which maps input to output.

2.1 Two Key Features of ML Models

Consider the best approximation problem, where, given a normed vector space Y of functions, a subspace $X \subset Y$, and a function $f \in Y$, we seek $\hat{\varphi} \in X$ satisfying

$$\forall \varphi \in X, \qquad \|f - \hat{\varphi}\|_b \leq \|f - \varphi\|_b; \tag{1}$$

if $\hat{\varphi} \in X$ exists, we call it the *best approximation* to $f \in Y$ from X with respect to the norm $\|\cdot\|_b$. We observe two prominent features of this problem. First, once they are given, the input parameters Y, X, and f do not change any more. Consequently, the question on the existence and uniqueness of the solution is answered unambiguously. Second, if the best approximation exists, the $\epsilon - \delta$ technique in mathematical analysis can usually be employed to design algorithms by which the best approximation can be computed with vanishing errors as we increase the computational resources.

Training an ML model can be regarded as coming up with a function that approximates the true function of the ML task, hence it bears some resemblance with the best approximation problem. However, these two problems are essentially different in that two key features

of training an ML model are almost opposite to those of the best approximation problem discussed in the previous paragraph.

(MLM-1) The data (and hence the problem) in ML are constantly evolving. Consequently, the performance of an ML model is not measured by how accurately it fits the data sets that are given *a priori*, but mostly by how it generalizes to *unseen* data sets.
(MLM-2) The data in ML always contain errors, which may even be very large for certain tasks. It is usually unnecessary and/or even impossible to gradually reduce the error to zero.

Due to (MLM-1,2), it is mandatory to analyze the effect of errors on the learning process, i.e., how model parameters change with respect to perturbations of the input data.

2.2 The Four Main Design Choices

As an ML strategy, an ERM method is specified by answers to four main design questions.

(ERM-1) What is the set of allowed functions from which we construct the predictor?
(ERM-2) How do we measure the performance of the predictor on the training data?
(ERM-3) How do we generalize the predictor so that it performs well on unseen test data?
(ERM-4) How to validate that the constructed predictor indeed minimizes the risk?

(ERM-1) has largely been answered for specific topics of ML. For example, in neural networks the set of allowed functions consists of linear maps and nonlinear activation functions such as the threshold function, the logistic function, and the rectified linear function.

Suppose in the setting of supervised learning that we have trained a predictor f from the set of given data $(\mathrm{x}_i, y_i)_{i=1}^N$ where y_i is the scalar label of the example x_i. Then (ERM-2) is answered by the *averaged loss*

$$\mathcal{L}_{\mathrm{ERM}} := \frac{1}{N}\sum_{i=1}^{N} \ell(y_i, \hat{y}_i), \tag{2}$$

where $\hat{y}_i := f(\mathrm{x}_i, \mathrm{w})$ is the output of the predictor for the ith example and w the vector of learnable parameters of f. In the training stage, we aim to find a good vector w^* of parameters such that the averaged loss is small; an example of ℓ that connects ERM to traditional numerical analysis is the least-square loss, where $\ell(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2$.

To see why ERM is named so, we can formally interpret w^* as the best parameter for the predictor f that minimizes the *expected risk*

$$\mathcal{L}_{\mathrm{exact}}(f) = \mathbb{E}_{\{\mathrm{x},y\}}[\ell(y, f(\mathrm{x}, \mathrm{w}))], \tag{3}$$

where $\mathbb{E}[\cdot]$ denotes the expectation, the subscript 'exact' emphasizes the fact that x, y must be taken over the (infinite) set of all possible data and labels. Since any ML model that runs on a computer must be trained by a *finite* amount of data, we are forced to approximate the

expected risk with an *empirical risk* for which x, y in the RHS of (3) are only taken over the training data.

Replacing the infinite set in (3) with a finite set gives rise to a potential pitfall called "overfitting" (the data), in which the training data are fitted very well yet the predictor performs poorly on unseen tests; see the last paragraph in Section 3.2. To prevent overfitting, we add an additional term to the averaged loss to obtain the regularized loss,

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{ERM}} + \alpha\Omega(\cdot), \tag{4}$$

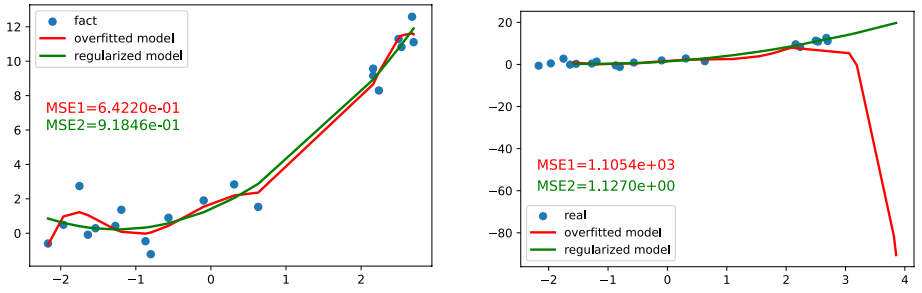
where $\alpha\Omega(\cdot)$ is the regularization term or the penalty term and $\alpha > 0$ is a hyperparameter. (4) is also an answer to (ERM-3). In Figure 1, we show an example of overfitting and regularization, where the penalty term effectively alleviates overfitting and improves the generalization of the ML model.

Given its critical role in ML, the generalization performance of a model should be quantitatively evaluated, which is usually done by choosing an appropriate function called the *evaluation metric*. Of course, the evaluation metric should cater for the specific ML task at hand and thus the same ML model might have different evaluation metrics for different ML tasks.

For (ERM-4), we can compare, using appropriate evaluation metrics, the performance of a number of different models constructed from the same set of functions in (ERM-1). There exist well-established techniques such as cross-validation; we will not delve into (ERM-4) any further.

3 Mathematical Aspects of Regularization

The fast growing literature on regularization calls for general and coherent frameworks that foster structured perspectives and systematic viewpoints to help our understanding of the large body of current research. In real-world applications, nonlinear models often work better than linear models. However, nonlinear phenomena are notoriously difficult to be unified theoretically; in most scenarios the analysis has to be performed on a case-by-case basis. Fortunately, a nonlinear process can be well approximated by a suitable sequence of linear



(a) training (b) test

Fig. 1 The blue dots, the red line, and the green line represent true values, an overfitted model, and a regularized model, respectively. MSE1 and MSE2 denote MSEs of the overfitted model and the regularized model, respectively

processes; this principle is ubiquitous in mathematics and we believe that it also works in ML.

In this section, we examine spectral cutoff and Tikhonov regularization, connecting them to well-known concepts in linear algebra. We choose these topics because they are analytically transparent and capture core design tensions (ill-posedness, bias-variance trade-offs, and consistency) upon which modern loss-based methods build. Furthermore, the SVD perspective (spectral cutoff) and the penalized least-squares perspective (Tikhonov) serve as paradigms whose proof patterns (such as discrepancy principle, degrees of freedom, and MSE decomposition) lead to strong heuristics in the nonlinear settings in Section 4. By focusing on these two linear perspectives, we aim to simplify complex phenomena, trace arrays of regularization methods to axioms of inner products on linear spaces, and reduce the entropy of the large body of regularization literature.

In the first four subsections, the core ideas of spectral cutoff and Tikhonov regularization, are fully explained under linear systems, using only basic concepts in linear algebra and numerical analysis. In Section 3.5, we show that Tikhonov regularization improves generalization by reducing the MSEs. For the design of regularized models, the consistency principle discussed in Section 3.6 should be satisfied, whether the systems are linear or nonlinear. Based on the convergence of regularization for linear systems, we propose a set of sufficient conditions that guarantee the consistency of both linear and nonlinear regularization methods. These conditions may be helpful in the design of new regularization methods.

3.1 Solvability of Linear Systems

The mathematical setting for our discussion of regularization is an inner-product space with the adjoint of a matrix $A \in \mathbb{C}^{m\times n}$ denoted by A^*, i.e.,

$$\forall \mathrm{x} \in \mathbb{C}^n,\ \forall \mathrm{y} \in \mathbb{C}^m,\ \langle A\mathrm{x}, \mathrm{y}\rangle = \langle \mathrm{x}, A^*\mathrm{y}\rangle . \tag{5}$$

Since A^*A is Hermitian and positive semi-definite, each eigenvalue of A^*A is non-negative. A *singular value* of $A \in \mathbb{C}^{m\times n}$, denoted by σ_j, is the non-negative square root of an eigenvalue of A^*A. The *singular value decomposition* (SVD) of a matrix $A \in \mathbb{C}^{m\times n}$ with rank r is a decomposition of the form $A = V\Lambda U^*$, where Λ is a diagonal matrix whose entries contain all singular values of A in non-increasing order,

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_n = 0,$$

$U = [\mathrm{u}_1, \mathrm{u}_2, \ldots, \mathrm{u}_n]$ and $V = [\mathrm{v}_1, \mathrm{v}_2, \ldots, \mathrm{v}_m]$ are square unitary matrices satisfying $U^*U = I$ and $V^*V = I$, and u_j's and v_j's are respectively the *right singular vectors* and the *left singular vectors* of A, i.e.,

$$\forall j = 1, 2, \ldots, r, \quad \begin{cases} A\mathrm{u}_j = \sigma_j \mathrm{v}_j; \\ A^*\mathrm{v}_j = \sigma_j \mathrm{u}_j, \end{cases} \tag{6a}$$

$$\begin{cases} \forall j = r+1, r+2, \ldots, n, & A\mathrm{u}_j = 0; \\ \forall j = r+1, r+2, \ldots, m, & A^*\mathrm{v}_j = 0. \end{cases} \tag{6b}$$

We also denote the SVD of a matrix A by $(\sigma_j, \mathrm{u}_j, \mathrm{v}_j)$.

Lemma 1 *(Solvability of a linear system) A linear system* $A\mathrm{x} = \mathrm{b}$ *with* $A \in \mathbb{C}^{m\times n}$ *admits a solution if and only if* b *is in the range (or image) of A, or equivalently,* b *is perpendicular to the null space of the adjoint of A, i.e.,*

$$\forall \mathrm{z} \in \{\mathrm{y} \in \mathbb{C}^m : A^*\mathrm{y} = 0\}, \quad \langle \mathrm{b}, \mathrm{z}\rangle = 0, \tag{7}$$

in which case a particular solution of the linear system is

$$\mathrm{x}^{\oplus} := \sum_{j=1}^{r} \frac{1}{\sigma_j} \langle \mathrm{b}, \mathrm{v}_j\rangle \, \mathrm{u}_j, \tag{8}$$

where $\langle\cdot,\cdot\rangle$ *denotes an inner product, r the rank of A, and* $(\sigma_j, \mathrm{u}_j, \mathrm{v}_j)$ *the SVD of A.*

Proof Suppose the linear system has a solution x. Then

$$\langle \mathrm{b}, \mathrm{z}\rangle = \langle A\mathrm{x}, \mathrm{z}\rangle = \langle \mathrm{x}, A^*\mathrm{z}\rangle = 0.$$

As for the sufficiency, suppose (7) holds. Then we have

$$\mathrm{b} = \sum_{j=1}^{r} \langle \mathrm{b}, \mathrm{v}_j\rangle \mathrm{v}_j = \sum_{j=1}^{r} \frac{1}{\sigma_j} \langle \mathrm{b}, \mathrm{v}_j\rangle A\mathrm{u}_j = A\mathrm{x}^{\oplus},$$

where the above steps follows from (6b), (6a), and (8). □

3.2 Ill-posed Linear Systems

The *condition number* of a function measures how much the output of the function can change for a small change in the input. For a linear system $A\mathrm{x} = \mathrm{b}$, the condition number based on the 2-norm is given by the singular values of A. If we perturb b to $\hat{\mathrm{b}} = \mathrm{b} + \epsilon \mathrm{v}_j$ for some $\epsilon \in \mathbb{C}$, by Lemma 1 the solution changes from x to $\hat{\mathrm{x}} = \mathrm{x} + \frac{\epsilon}{\sigma_j}\mathrm{u}_j$. Then the condition number with respect to *absolute* errors is

$$\frac{\|\hat{\mathrm{x}} - \mathrm{x}\|_2}{\|\hat{\mathrm{b}} - \mathrm{b}\|_2} = \frac{1}{\sigma_j}. \tag{9}$$

Thus a solvable linear system is ill-conditioned if its matrix has small singular values.

While our derivations only model perturbations in the right-hand side b for clarity, many applications contain errors in both A and b. In that case, total least squares (TLS) leads to regularizers that cover the augmented perturbation $[A \,|\, \mathrm{b}] \mapsto [A + \Delta A \,|\, \mathrm{b} + \Delta \mathrm{b}]$. Our linear results do not require the TLS machinery, but the guidance on choosing the regularization strength (discrepancy principles, cross-validation/generalized cross-validation (GCV) paths) remains applicable when perturbations in A are not negligible.

Following Hadamard (1923), we call a problem *well-posed* if (i) it admits a solution, (ii) the solution is unique, and (iii) the problem has a small condition number; otherwise we say the problem is *ill-posed*. Hereafter we consider typical cases of ill-posed linear systems.

Theorem 2 *Suppose the linear system* $A\mathrm{x} = \mathrm{b}$ *in Lemma 1 has no solutions. Then* $\mathrm{x}^{\oplus}$ *in (8) solves the problem of minimizing the 2-norm of the residual, i.e.,*

$$\mathrm{x}^{\oplus} = \arg\min_{\mathrm{x}\in\mathbb{C}^n} \|A\mathrm{x} - \mathrm{b}\|_2. \tag{10}$$

Proof By Lemma 1, the right-hand side (RHS) b is not in the range of A, but we must have $\mathrm{b} \in \mathbb{C}^m$ and consequently $\mathrm{b} = \sum_{i=1}^{m} \langle \mathrm{b}, \mathrm{v}_j \rangle \mathrm{v}_j$. Then (8) yields

$$\forall \mathrm{b} \notin \mathrm{range} A, \quad A\mathrm{x}^{\oplus} - \mathrm{b} \in \mathrm{span}(\mathrm{v}_{r+1}, \mathrm{v}_{r+2}, \dots, \mathrm{v}_m).$$

On the other hand, the rank of A being r implies

$$\forall \mathrm{x} \in \mathbb{C}^n, \quad A\mathrm{x} - A\mathrm{x}^{\oplus} \in \mathrm{span}(\mathrm{v}_1, \mathrm{v}_2, \dots, \mathrm{v}_r).$$

Hence $\langle A\mathrm{x}^{\oplus} - \mathrm{b}, A\mathrm{x} - A\mathrm{x}^{\oplus} \rangle = 0$ and the Pythagorean theorem yields

$$\|A\mathrm{x} - \mathrm{b}\|_2^2 = \|A\mathrm{x} - A\mathrm{x}^{\oplus}\|_2^2 + \|A\mathrm{x}^{\oplus} - \mathrm{b}\|_2^2,$$

which completes the proof. □

In ML, Theorem 2 corresponds to the fortunate case where the amount of available data is greater than the number of parameters that determine the predictor.

Theorem 3 *Suppose a linear system* $A\mathrm{x} = \mathrm{b}$ *has multiple solutions. Then its general solution is* $\mathrm{x}^{\oplus} + \mathrm{y}$ *where* $\mathrm{x}^{\oplus}$ *is given by (8) and* y *is any vector in the null space of A. Furthermore,* $\mathrm{x}^{\oplus}$ *is the solution with the minimum 2-norm.*

Proof Lemma 1 implies $\mathrm{x}^{\oplus} \in \mathrm{span}(\mathrm{u}_1, \mathrm{u}_2, \dots, \mathrm{u}_r)$ and $\mathrm{null}\, A = \mathrm{span}(\mathrm{u}_{r+1}, \mathrm{u}_{r+2}, \dots, \mathrm{u}_n)$. Hence $\forall \mathrm{y} \in \mathrm{null}\, A$, $\langle \mathrm{x}^{\oplus}, \mathrm{y} \rangle = 0$ and the Pythagorean theorem yields

$$\|\mathrm{x}^{\oplus} + \mathrm{y}\|_2 = \|\mathrm{x}^{\oplus}\|_2 + \|\mathrm{y}\|_2.$$

The proof is completed by the fact that all solutions of the linear system are of the form $\mathrm{x}^{\oplus} + \mathrm{y}$. □

By the uniqueness of $\mathrm{x}^{\oplus}$ in Theorems 2 and 3, we can define, with respect to minimizing the 2-norm, a unique solution for a linear system that either has no solution or has multiple solutions. This leads to the *Moore-Penrose inverse* or *pseudo-inverse* $A^+ \in \mathbb{C}^{n\times m}$ of a matrix $A \in \mathbb{C}^{m\times n}$,

$$A^+\mathrm{y} = \sum_{j=1}^{r} \frac{1}{\sigma_j} \langle \mathrm{y}, \mathrm{v}_j \rangle \mathrm{u}_j, \tag{11}$$

where r is the rank of A and $(\sigma_j, \mathrm{u}_j, \mathrm{v}_j)$ denotes its SVD.

For the two typical cases in Lemma 4, the pseudo-inverse can be expressed in terms of A and A^*, with the left inverse corresponding to linear least squares and the right inverse to a linear system with multiple solutions.

Lemma 4 *If A has linearly independent columns, then A^*A is invertible and we have*

$$A^+ = (A^*A)^{-1}A^*, \tag{12}$$

which is then a left inverse, i.e. $A^+A = I$. Similarly, if A has linearly independent rows, then AA^* is invertible and

$$A^+ = A^*(AA^*)^{-1}, \tag{13}$$

which is then a right inverse, i.e. $AA^+ = I$.

Almost all ML tasks are ill-posed. As a common scenario, the given data sets are insufficient to determine the best predictor; this corresponds to the case of Theorem 3. As the amount of given data increases, the problem would eventually become a least square problem that corresponds to the ideal case of Theorem 2. Unfortunately, this strategy is often not feasible, because the extra data is either unavailable or too expensive to collect. From a mathematical viewpoint, the number of possible functions that can be used to construct the predictor is always greater than that of the given data; after all, the former is infinite while the latter is finite.

This ill-posedness of ML tasks can be partially overcome by supplementing the given data sets with a number of assumptions known as the *inductive bias* on the characteristics of the physical processes to be modeled. One typical example is (ERM-1) in Section 2.2 where the construction of the ML model is limited to a set of allowed functions.

Since both the given data sets and the inductive bias are exploited in ML, we should avoid two potential pitfalls. On one hand, if the inductive bias is too strong, or the set of allowable functions in (ERM-1) is too simple, the model would have a poor performance on the given data, which usually means a poor performance on the unseen data. This situation is called *underfitting* (the data) in that the predictor fails to capture salient features in the existing data. On the other hand, if the inductive bias is too weak, or the set of allowable functions in (ERM-1) is too complicated, the model would fit accurately to the given data but have a poor performance on the unseen data. This situation is called *overfitting* (the data), in which case the model essentially mistakes noises in the data as patterns of realistic processes.

3.3 Spectral Cutoff for Ill-Conditioning

The *spectral cutoff* stabilizes an ill-conditioned linear system by neglecting in (8) the terms with small singular values. But exactly which terms are to be dropped? As a key observation, the linear system only needs to be satisfied up to the accuracy of the RHS vector.

This observation leads to

Definition 1 Given a perturbed RHS $\hat{\mathrm{b}}$ to a linear system $A\mathrm{x} = \mathrm{b}$, the *discrepancy principle* is a strategy to come up with an approximate solution $\hat{\mathrm{x}}$ so that

$$\|A\hat{\mathrm{x}} - \hat{\mathrm{b}}\|_2 = C\|\mathrm{b} - \hat{\mathrm{b}}\|_2, \tag{14}$$

where $C \geq 1$ is some suitable constant.
For the spectral cutoff, the discrepancy principle guides us to approximate (8) by

$$\mathrm{x}_p := \begin{cases} 0 & \text{if } p = 0; \\ \sum_{j=1}^{p} \frac{1}{\sigma_j} \left\langle \hat{\mathrm{b}}, \mathrm{v}_j \right\rangle \mathrm{u}_j & \text{if } p = 1, \ldots, r, \end{cases} \tag{15}$$

where p is determined by the perturbation error $\|\hat{\mathrm{b}} - \mathrm{b}\|_2$ so that (14) is satisfied. A common choice is $C = 1$. The questions that remain to be answered are: is the existence of p guaranteed? exactly how is p determined?

Theorem 5 *For $A \in \mathbb{C}^{m\times n}$, suppose the linear system $A\mathrm{x} = \mathrm{b}$ is solvable and a perturbation $\hat{\mathrm{b}}$ to b satisfies*

$$\|\hat{\mathrm{b}} - \mathrm{b}\|_2 \leq \epsilon \leq \|\hat{\mathrm{b}}\|_2. \tag{16}$$

Then there exists a smallest integer $p = p(\epsilon)$ such that

$$\|A\mathrm{x}_p - \hat{\mathrm{b}}\|_2 \leq \epsilon, \tag{17}$$

where x_p is defined in (15).

Proof The definition $f(p) := \|A\mathrm{x}_p - \hat{\mathrm{b}}\|_2^2 - \epsilon^2$ can be rewritten as

$$f(p) = \sum_{j=p+1}^{m} \left|\left\langle \hat{\mathrm{b}}, \mathrm{v}_j \right\rangle\right|^2 - \epsilon^2, \tag{18}$$

where v_j is a left singular vector of A. Then (16) yields $f(0) = \|\hat{\mathrm{b}}\|_2^2 - \epsilon^2 \geq 0$ and $f(r) = -\epsilon^2 < 0$ if the rank r of A equals m. If $r < m$, the solvability condition of the linear system and Lemma 1 yield $\left\langle \mathrm{b}, \mathrm{v}_j \right\rangle = 0$ for each $j = r+1, \cdots, m$. Then we have

$$f(r) = \sum_{j=r+1}^{m} \left|\left\langle (\hat{\mathrm{b}} - \mathrm{b}), \mathrm{v}_j \right\rangle\right|^2 - \epsilon^2 \leq \|\hat{\mathrm{b}} - \mathrm{b}\|_2^2 - \epsilon^2 \leq 0$$

where the last step follows from (16). By (18), f is monotonically decreasing, and the first conclusion concerning (17) follows from $f(0) \geq 0$ and $f(r) \leq 0$. □

Remark 1 In practical computation, p can be determined by calculating the sum in (15) until the RHS of (18) becomes zero or negative for the first time.

Spectral cutoff is closely related to the Occam's razor principle, which favors the simplest solution out of multiple models. The sparse regularization methods discussed in Section 5.2 can also be considered as analogous to spectral cutoff.

Corollary 6 *The discrepancy principle for the spectral cutoff is convergent in that*

$$\lim_{\epsilon \to 0} \mathrm{x}_p = A^+ \mathrm{b}. \tag{19}$$

Proof By (17), we have $\|A\mathrm{x}_p - \mathrm{b}\|_2 \leq \|A\mathrm{x}_p - \hat{\mathrm{b}}\|_2 + \|\hat{\mathrm{b}} - \mathrm{b}\|_2 \leq 2\epsilon$. Then Lemma 4 yields $A^+ A\mathrm{u} = \mathrm{u}$ for all u in $\mathrm{span}(\mathrm{u}_1, \ldots, \mathrm{u}_r)$, which gives (19). □

As a potential drawback of spectral cutoff, the knowledge of the full spectrum of A^*A is required, which might be computationally expensive. In comparison, Tikhonov regularization does not require any explicit knowledge of the SVD of A.

3.4 Tikhonov Regularization

Tikhonov regularization (Tikhonov and Arsenin, 1977) is a method to approximate the solution of an ill-conditioned linear system $A\mathrm{x} = \mathrm{b}$ by that of a regularized linear system

$$\alpha \mathrm{x} + A^* A\mathrm{x} = A^* \mathrm{b}, \tag{20}$$

where $\alpha > 0$ is called the *regularization parameter*.

Ridge estimation is a method to estimate parameters in a linear model by solving a regularized least squares problem. Specifically, it minimizes the sum of squared residuals with an added penalty on the 2-norm of the coefficient vector. The ridge estimator is defined as the solution to the optimization problem

$$\hat{\mathrm{x}}_\alpha = \mathrm{argmin}_x \left(\|A\mathrm{x} - \mathrm{b}\|^2 + \alpha \|\mathrm{x}\|^2 \right), \tag{21}$$

where b is the response vector, A is the design matrix, x is the vector of coefficients, and $\alpha > 0$ is the regularization parameter that controls the trade-off between the goodness of fit and the magnitude of the coefficients.

From classical work on ridge regression (Hoerl and Kennard, 1970) and regularization of ill-posed problems (Tikhonov and Arsenin, 1977; Hansen, 2010; Wahba, 1990), we *recall* the equivalence between ridge/Tikhonov and L_2-penalized least squares and include short proofs for the sake of completeness.

Theorem 7 *Suppose a matrix $A \in \mathbb{C}^{m \times n}$ has its SVD $(\sigma_j, \mathrm{u}_j, \mathrm{v}_j)$ with rank r. Then for any $\mathrm{b} \in \mathbb{C}^m$ the Tikhonov regularized linear system (20) admits the unique solution*

$$\mathrm{x}_\alpha := \sum_{j=1}^{r} \frac{\sigma_j}{\alpha + \sigma_j^2} \langle \mathrm{b}, \mathrm{v}_j \rangle \mathrm{u}_j \tag{22}$$

and satisfies $\lim_{\alpha \to 0} (\alpha I + A^* A)^{-1} A^* \mathrm{b} = A^+ \mathrm{b}$.

Proof The matrix $B := \alpha I + A^*A$ is strictly positive definite and thus nonsingular. Furthermore, $B\mathrm{u}_j = (\alpha + \sigma_j^2)\mathrm{u}_j$ and $B^* = B$ imply that $(\alpha + \sigma_j^2, \mathrm{u}_j, \mathrm{u}_j)$ is a singular system of B. Apply Lemma 1 to (20) and we have

$$\mathrm{x} = \sum_{j=1}^{r} \frac{1}{\alpha + \sigma_j^2} \langle A^*\mathrm{b}, \mathrm{u}_j \rangle \, \mathrm{u}_j = \sum_{j=1}^{r} \frac{1}{\alpha + \sigma_j^2} \langle \mathrm{b}, A\mathrm{u}_j \rangle \, \mathrm{u}_j = \mathrm{x}_\alpha,$$

where the last step is due to (6a) and (22). Finally, the limit follows from (11) and (22). □

Remark 2 Before turning to consistency and MSE trade-offs specific to our setup, we recall the standard equivalence between ridge regression and least squares with an L_2 penalty and provide a concise proof (Theorem 8) for completeness.

Theorem 8 *For any $\mathrm{b} \in \mathbb{C}^m$, the solution x_α in (22) to the Tikhonov regularized linear system (20) is equivalent to the unique minimizer of $\|A\mathrm{x} - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}\|_2^2$, i.e.,*

$$\|A\mathrm{x}_\alpha - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}_\alpha\|_2^2 = \inf_{\mathrm{x} \in \mathbb{C}^n} \left\{ \|A\mathrm{x} - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}\|_2^2 \right\}. \tag{23}$$

Proof Define $\mathrm{z}_\alpha := \alpha\mathrm{x}_\alpha + A^*A\mathrm{x}_\alpha - A^*\mathrm{b}$ and we know from the law of cosines that

$$\begin{aligned} &\|A\mathrm{x} - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}\|_2^2 = 2\mathrm{Re}\, \langle \mathrm{x} - \mathrm{x}_\alpha, \mathrm{z}_\alpha \rangle \\ &+ \|A\mathrm{x}_\alpha - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}_\alpha\|_2^2 + \|A\mathrm{x} - A\mathrm{x}_\alpha\|_2^2 + \alpha\|\mathrm{x} - \mathrm{x}_\alpha\|_2^2. \end{aligned} \tag{24}$$

Hence the solution x_α of (20) satisfies $\mathrm{z}_\alpha = 0$ and consequently (23) holds.

Conversely, let x_α be the solution of (23) and suppose $\mathrm{z}_\alpha \neq 0$. Then for $\mathrm{x} := \mathrm{x}_\alpha - \epsilon\mathrm{z}_\alpha$ with $\epsilon \in \mathbb{R}$, (24) yields

$$\|A\mathrm{x} - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}\|_2^2 = \|A\mathrm{x}_\alpha - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}_\alpha\|_2^2 - 2\epsilon a + \epsilon^2 b,$$

where $a := \|\mathrm{z}_\alpha\|_2^2$ and $b := \|A\mathrm{z}_\alpha\|_2^2 + \alpha\|\mathrm{z}_\alpha\|_2^2$. The choice of $\epsilon = \dfrac{a}{b}$ gives

$$\|A\mathrm{x} - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}\|_2^2 < \|A\mathrm{x}_\alpha - \mathrm{b}\|_2^2 + \alpha\|\mathrm{x}_\alpha\|_2^2,$$

which contradicts (23) and completes the proof. □

Remark 3 By Theorem 8, Tikhonov regularization keeps the residual $\|A\mathrm{x}_\alpha - \mathrm{b}\|_2$ small and simultaneously prevents $\|\mathrm{x}_\alpha\|_2$ from being too large. This is an important observation for ML, as x_α are often the model parameters and the main point of regularization is to keep $\|\mathrm{x}_\alpha\|_2$ small through the penalty term for good generalization performance.

A comparison of (23) to (4) shows that for linear systems we have $\Omega = \|\mathrm{x}_\alpha\|$ in (4). As for the determination of the hyperparameter α in (4), we defer its discussion to after Theorem 9.

As noted in (MLM-2) of Section 2.1, the RHS b of a linear system often has a certain degree of errors, say, $\|\hat{\mathrm{b}} - \mathrm{b}\|_2 < \epsilon$. In this case the exact solution $\mathrm{x} = A^+\mathrm{b}$ is approximated by x_α that satisfies $B\mathrm{x}_\alpha = A^*\hat{\mathrm{b}}$, where $B := \alpha I + A^*A$. Hence the total absolute error is

$$\mathrm{x}_\alpha - \mathrm{x} = B^{-1}A^*(\hat{\mathrm{b}} - \mathrm{b}) + (B^{-1}A^*\mathrm{b} - A^+\mathrm{b}), \tag{25}$$

where the first RHS term is the error caused by the perturbation to the data b and the second that caused by the regularization. On one hand, α should be small to keep the regularization error small. On the other hand, we know from basic linear algebra that the condition number of the matrix B is

$$\mathrm{cond}_2 B = \frac{\alpha + \sigma_1^2}{\alpha + \sigma_n^2}. \tag{26}$$

Thus α should be large to make the data error small; otherwise the conditioning of the linear system $B\mathrm{x} = A^*(\hat{\mathrm{b}} - \mathrm{b})$ would be large; the regularization effect is clearly shown by comparing (26) to (9). To sum up, the value of α should be determined via a compromise between accuracy and stability.

The principle of balancing accuracy and stability should also be followed in designing ML models; otherwise two undesirable results may occur: in the case of overfitting we have good accuracy for the training data, but the model is not stable for unseen data; in the case of underfitting we have good stability for unseen data, but the accuracy may be unsatisfactory for both training and unseen data.

Naturally the next step is: how to strike the balance *quantitatively*? Both this question and that on the existence and uniqueness of α raised after Theorem 7 can be answered by once again resorting to the discrepancy principle.

Theorem 9 *Suppose the linear system* $A\mathrm{x} = \mathrm{b}$ *with* $A \in \mathbb{C}^{m\times n}$ *is solvable and a perturbation* $\hat{\mathrm{b}}$ *to* b *satisfies*

$$\|\hat{\mathrm{b}} - \mathrm{b}\|_2 \le \epsilon \le \|\hat{\mathrm{b}}\|_2. \tag{27}$$

Then there exists a unique $\alpha = \alpha(\epsilon)$ *such that*

$$\|A\mathrm{x}_\alpha - \hat{\mathrm{b}}\|_2 = \epsilon, \tag{28}$$

where x_α *is the unique solution of*

$$(\alpha I + A^*A)\mathrm{x} = A^*\hat{\mathrm{b}}. \tag{29}$$

Proof By (22), $f(\alpha) := \|A\mathrm{x}_\alpha - \hat{\mathrm{b}}\|_2^2 - \epsilon^2$ is the same as

$$f(\alpha) = \sum_{j=1}^{m} \left| \frac{\alpha^2}{(\alpha + \sigma_j^2)^2} \left\langle \hat{\mathrm{b}}, \mathrm{v}_j \right\rangle \right|^2 - \epsilon^2,$$

where v_j is a left singular vector of A. Being continuous and strictly monotonically increasing, f also satisfies

$$\lim_{\alpha\to 0} f(\alpha) = -\epsilon^2 < 0, \quad \lim_{\alpha\to+\infty} f(\alpha) = \|\hat{\mathrm{b}}\|_2^2 - \epsilon^2 \geq 0,$$

hence f must have exactly one root α that satisfies (28). □

Remark 4 In practice, the parameter α is seldom obtained from (28). Instead, we start from some α with $f(\alpha) > 0$ and repeatedly decrease it by a constant factor until $f(\alpha)$ turns negative.

According to Theorem 9, the best balance between accuracy and stability can be achieved by setting (28), which is a manifestation of the discrepancy principle in Definition 1. Despite that the errors of training data are sometimes unknown, Theorem 9 is still a good guide to determine the value of the hyperparameter in (4).

Corollary 10 *The discrepancy principle for Tikhonov regularization is convergent, i.e.,* $\lim_{\epsilon\to 0} \mathrm{x}_\alpha = A^+\mathrm{b}$.

Proof (28) and the triangle inequality yield

$$\|\hat{\mathrm{b}}\|_2 - \epsilon = \|\hat{\mathrm{b}}\|_2 - \|A\mathrm{x}_\alpha - \hat{\mathrm{b}}\|_2 \leq \|A\mathrm{x}_\alpha\|_2.$$

On the other hand, x_α being a solution of (29) gives

$$\alpha\|A\mathrm{x}_\alpha\|_2 = \|AA^*(\hat{\mathrm{b}} - A\mathrm{x}_\alpha)\|_2 \leq \|AA^*\|_2\epsilon.$$

(27) implies $\|\hat{\mathrm{b}}\|_2 \geq \|\mathrm{b}\|_2 - \epsilon$, which, together with the above two inequalities, gives $\alpha \leq \frac{\|AA^*\|_2\epsilon}{\|\mathrm{b}\|_2 - 2\epsilon}$. Hence $\lim_{\epsilon\to 0} \alpha = 0$. Then the conclusion follows from (11) and (22). □

3.5 MSEs Reduced by Tikhonov Regularization

We show that Tikhonov regularization improves the generalization performance of a linear regression model,

$$\boldsymbol{y} = X\beta + \boldsymbol{\epsilon}, \tag{30}$$

where $X \in \mathbb{R}^{n\times d}$ and $\boldsymbol{y} \in \mathbb{R}^{n\times 1}$ are the training data and their corresponding labels, respectively; $\beta \in \mathbb{R}^{d\times 1}$ contains the model parameters and $\boldsymbol{\epsilon}$ is the vector of unobserved random perturbations. Without loss of generality, we assume that X has full column rank so that X^TX is invertible. We also assume zero expectation and joint independence, i.e.,

$$\mathbb{E}(\boldsymbol{\epsilon}) = 0; \quad \mathrm{Cov}(\boldsymbol{\epsilon}) = \sigma^2 I, \tag{31}$$

where $\sigma > 0$ and I is the identity matrix. In the following, we measure the generalization performance by MSEs. More detailed information can be found in Farebrother (1976).

Lemma 11 *For a test datum $\boldsymbol{x}_0 \in \mathbb{R}^d$, the least square estimator (LSE) given by Theorem 2 and Lemma 4,*

$$\hat{\boldsymbol{\beta}} := (X^T X)^{-1} X^T \boldsymbol{y}, \tag{32}$$

predicts $\hat{y}_0 := \boldsymbol{x}_0^T \hat{\boldsymbol{\beta}}$ that satisfies

$$\begin{cases} \mathbb{E}(\hat{y}_0) &= \boldsymbol{x}_0^T \boldsymbol{\beta}, \\ \mathrm{Var}(\hat{y}_0) &= \sigma^2 \boldsymbol{x}_0^T (X^T X)^{-1} \boldsymbol{x}_0, \\ \mathrm{MSE}(\hat{y}_0) &= \sigma^2 \boldsymbol{x}_0^T (X^T X)^{-1} \boldsymbol{x}_0 + \sigma^2. \end{cases} \tag{33}$$

Proof (32) and (30) imply $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (X^T X)^{-1} X^T \boldsymbol{\epsilon}$. Then (31) and basic identities on covariance matrices yield

$$\begin{aligned} \mathbb{E}(\hat{y}_0) &= \mathbb{E}(\boldsymbol{x}_0^T \boldsymbol{\beta} + \boldsymbol{x}_0^T (X^T X)^{-1} X^T \boldsymbol{\epsilon}) \\ &= \boldsymbol{x}_0^T \boldsymbol{\beta} + \boldsymbol{x}_0^T (X^T X)^{-1} X^T \mathbb{E}(\boldsymbol{\epsilon}) = \boldsymbol{x}_0^T \boldsymbol{\beta}, \\ \mathrm{Var}(\hat{y}_0) &= \mathrm{Var}(\boldsymbol{x}_0^T \boldsymbol{\beta} + \boldsymbol{x}_0^T (X^T X)^{-1} X^T \boldsymbol{\epsilon}) \\ &= \mathrm{Var}(\boldsymbol{x}_0^T (X^T X)^{-1} X^T \boldsymbol{\epsilon}) \\ &= \boldsymbol{x}_0^T (X^T X)^{-1} X^T \mathrm{Cov}(\boldsymbol{\epsilon}) X (X^T X)^{-1} \boldsymbol{x}_0 \\ &= \sigma^2 \boldsymbol{x}_0^T (X^T X)^{-1} \boldsymbol{x}_0. \end{aligned}$$

By (31), an observed sample $y_0 := \boldsymbol{x}_0^T \boldsymbol{\beta} + \epsilon_0$ satisfies both $\mathbb{E}(\epsilon_0) = 0$ and $\mathrm{Var}(\epsilon_0) = \sigma^2$. Then

$$\begin{aligned} \mathrm{MSE}(\hat{y}_0) &= \mathbb{E}(\hat{y}_0 - y_0)^2 = \mathbb{E}(\hat{y}_0 - \boldsymbol{x}_0^T \boldsymbol{\beta} - \epsilon_0)^2 \\ &= \mathbb{E}(\hat{y}_0 - \boldsymbol{x}_0^T \boldsymbol{\beta})^2 + \mathbb{E}(\epsilon_0^2) = \mathbb{E}[\hat{y}_0 - \mathbb{E}(\hat{y}_0)]^2 + \sigma^2 \\ &= \mathrm{Var}(\hat{y}_0) + \sigma^2, \end{aligned}$$

where the independence of $\boldsymbol{\epsilon}$ and ϵ_0 gives the third step. □

If the matrix X of training data has small singular values, the prediction by LSE (32) has large MSEs, even though it is not biased; this is often caused by the (almost) multicollinearity of X. Tikhonov regularization is an effective solution of this problem.

Lemma 12 *For a test datum $\boldsymbol{x}_0 \in \mathbb{R}^d$, the Tikhonov regularization estimator given by (20) and Theorem 8,*

$$\hat{\boldsymbol{\beta}}_\alpha := (\alpha I + X^T X)^{-1} X^T \boldsymbol{y}, \tag{34}$$

predicts $\hat{y}_\alpha := \boldsymbol{x}_0^T \hat{\boldsymbol{\beta}}_\alpha$ that satisfies

$$\begin{cases} \mathbb{E}(\hat{y}_\alpha) = & \boldsymbol{x}_0^T (I - \alpha H) \boldsymbol{\beta}, \\ \mathrm{Var}(\hat{y}_\alpha) = & \sigma^2 \boldsymbol{x}_0^T H X^T X H \boldsymbol{x}_0, \\ \mathrm{MSE}(\hat{y}_\alpha) = & \boldsymbol{x}_0^T H (\alpha^2 \boldsymbol{\beta} \boldsymbol{\beta}^T + \sigma^2 X^T X) H \boldsymbol{x}_0 + \sigma^2, \end{cases} \tag{35}$$

where $H := (\alpha I + X^T X)^{-1}$ is a shorthand notation.

Proof The proof is similar to that of Lemma 11. □

The above lemmas lead to the main conclusion in Theorem 13 that *Tikhonov regularization can improve the generalization performance of the linear regression model* (30). Furthermore, in terms of MSEs this performance improvement can be maximized by suitable values of the hyperparameter α.

Theorem 13 *Characterize the trained model in (30) by*

$$B := \sigma^2(X^TX)^{-1} - \beta\beta^T \tag{36}$$

and let $\lambda_{\min}(B)$ *denote the smallest eigenvalue of B. If* $\lambda_{\min}(B) \geq 0$*, then* $\hat{y}_\alpha$ *is always better than* $\hat{y}_0$*, i.e.,*

$$\forall \alpha > 0, \forall \boldsymbol{x}_0 \neq 0, \quad \text{MSE}(\hat{y}_\alpha) < \text{MSE}(\hat{y}_0); \tag{37}$$

otherwise, $\hat{y}_\alpha$ *is better than* $\hat{y}_0$ *when* $0 < \alpha < 2\sigma^2/|\lambda|$*, i.e.*

$$\forall \alpha \in (0, 2\sigma^2/|\lambda|), \forall \boldsymbol{x}_0 \neq 0, \quad \text{MSE}(\hat{y}_\alpha) < \text{MSE}(\hat{y}_0). \tag{38}$$

Furthermore, $\text{MSE}(\hat{y}_\alpha)$ *is minimized if and only if we maximize* $\dfrac{\alpha^2\lambda + 2\alpha\sigma^2}{(\alpha+\mu)^2}$

Proof Using the shorthand H in Lemma 12, we have

$$\begin{aligned} & \text{MSE}(\hat{y}_0) - \sigma^2 = \sigma^2 \boldsymbol{x}_0^T (X^TX)^{-1} \boldsymbol{x}_0 \\ = \;& \sigma^2 \boldsymbol{x}_0^T H H^{-1} (X^TX)^{-1} \boldsymbol{x}_0 \\ = \;& \sigma^2 \boldsymbol{x}_0^T H [\alpha (X^TX)^{-1} + I] H^{-1} H \boldsymbol{x}_0 \\ = \;& \sigma^2 \boldsymbol{x}_0^T H [\alpha^2 (X^TX)^{-1} + 2\alpha I + X^TX] H \boldsymbol{x}_0. \end{aligned}$$

Then the symmetry of H and B implies

$$\begin{aligned} \text{MSE}(\hat{y}_0) - \text{MSE}(\hat{y}_\alpha) &= \alpha \boldsymbol{x}_0^T H [2\sigma^2 I + \alpha B] H \boldsymbol{x}_0 \\ &\geq \alpha [2\sigma^2 + \alpha \lambda_{\min}(B)] \|H\boldsymbol{x}_0\|_2^2, \end{aligned}$$

where the last step follows from $\boldsymbol{\xi}^T B \boldsymbol{\xi} \geq \lambda_{\min}(B)\|\boldsymbol{\xi}\|_2^2$ and the equality holds when $\boldsymbol{\xi}$ is an eigenvector of B for $\lambda_{\min}(B)$. Therefore, both (37) and (38) hold. Since no eigenvalue of $H^{-1} = \alpha I + X^TX$ is less than $\alpha + \mu$, all eigenvalues of H are in $(0, \dfrac{1}{\alpha+\mu}]$, indicating $\|H\boldsymbol{x}_0\|_2 \leq \dfrac{1}{\alpha+\mu}\|\boldsymbol{x}_0\|_2$. Then

$$\alpha(2\sigma^2 + \alpha\lambda)\|H\boldsymbol{x}_0\|_2^2 \leq \frac{\alpha^2\lambda + 2\alpha\sigma^2}{(\alpha+\mu)^2}\|\boldsymbol{x}_0\|_2^2$$

and the last conclusion holds. □

Choosing the ridge parameter α without a noise estimate

Setup. For $y = X\beta^\star + \varepsilon$ and the ridge/Tikhonov smoother $\hat{y}_\alpha = S_\alpha y$ with $S_\alpha = X(X^\top X + \alpha I)^{-1}X^\top$, *GCV* chooses α by minimizing

$$\mathrm{GCV}(\alpha) = \frac{\|(I - S_\alpha)y\|_2^2}{\left(1 - \frac{1}{n}\operatorname{tr}(S_\alpha)\right)^2},$$

which *does not* require any noise-variance estimate (Hansen (2010), Wahba (1990), Golub et al. (1979), Craven and Wahba (1979)). In the case of a thin SVD with $X = U\Sigma V^\top$, we have

$$\hat{y}_\alpha = U\,\mathrm{diag}\left(\frac{\sigma_i^2}{\sigma_i^2+\alpha}\right)U^\top y, \qquad \operatorname{tr}(S_\alpha) = \sum_{i=1}^{r}\frac{\sigma_i^2}{\sigma_i^2+\alpha}.$$

Workflow. Compute a (thin or randomized) SVD once; scan α on a log grid (e.g., 10^{-6}–10^{3}); pick $\hat{\alpha} = \arg\min_\alpha \mathrm{GCV}(\alpha)$; optionally verify on 5-fold cross-validation and, if minima differ by $> 10\times$, use their geometric mean. For large n, approximate $\operatorname{tr}(S_\alpha)$ with Hutchinson's estimator $\frac{1}{m}\sum_{j=1}^{m} z_j^\top S_\alpha z_j$ with Rademacher z_j. This turns the qualitative bias–variance discussion into a single-pass workflow.

Improving the adversarial robustness and interpretability of deep neural networks by investigating their input gradients In addition, the L-curve method is proposed in Hansen and O'Leary (1993); Hansen (2001) to determine the regularization parameter, which is shown to be more robust in the presence of correlated errors.

3.6 The Consistency Principle as a Guideline for the Design of Regularized Models

Now, we consider nonlinear systems and focus on the consistency of the regularizer. It is fundamentally important that a regularized model be consistent with the original model in the sense of

Definition 2 *(Consistency principle)*

Consider an ML task $\boldsymbol{y} = f(X, \beta)$, where $X \in \mathbb{R}^{n\times m}$ and $\boldsymbol{y} \in \mathbb{R}^n$ are respectively the training data and the corresponding labels and $\beta \in \mathbb{R}^d$ are the model parameters. Regularizing the loss function $\mathcal{L}(\beta)$ with an additional term $\alpha\Omega(\beta)$ yields

$$\hat{\beta}_0 = \mathrm{argmin}_{\beta\in\mathbb{R}^d}\,\mathcal{L}(\beta), \tag{39a}$$

$$\hat{\beta}_\alpha = \mathrm{argmin}_{\beta\in\mathbb{R}^d}[\mathcal{L}(\beta) + \alpha\Omega(\beta)]. \tag{39b}$$

The regularized model is *consistent* with the original one if

$$\lim_{\alpha\to 0^+}\hat{\beta}_\alpha = \hat{\beta}_0. \tag{40}$$

Under what conditions can we guarantee that a regularization method satisfies this consistency principle?

From Definition 2, one might argue that, because $\lim_{\alpha\to 0^+} \mathcal{L}(\beta) + \alpha\Omega(\beta) = \mathcal{L}(\beta)$, the consistency condition (40) follows trivially from (39); but this argument is incorrect. Consider the counterexample,

$$\hat{\beta}_0 = \text{argmin}_{x\in\mathbb{R}} \frac{|x|}{1+x^2}, \tag{41a}$$

$$\hat{\beta}_\alpha = \text{argmin}_{x\in\mathbb{R}} \left[\frac{|x|}{1+x^2} + \alpha\Big(|x| + \frac{1}{|x|}\Big)\right], \tag{41b}$$

where $\Omega(x) := |x| + \frac{1}{|x|}$ is chosen so that $|x|$ is neither too large nor too small. The solutioin of (41a) is clearly $\hat{\beta}_0 = 0$ while that of (41b) follows from the inequality $a + b \geq 2\sqrt{ab}$ for $a, b \geq 0$, i.e., $\hat{\beta}_\alpha$ is the root of $|x| + \frac{1}{|x|} = \frac{1}{\sqrt{\alpha}}$ and might tend to infinity as $\alpha \to 0$. In a word, (40) does not follow directly from (39).

For linear systems, we have already shown in Corollaries 6 and 10 that both the spectral cutoff and the Tikhonov regularization are convergent and thus indeed consistent. Is it possible to generalize the results to a set of principles that could also guide regularization of nonlinear systems?

In this subsection, we give a positive answer to the above question. After recalling some well-known theorems in real analysis, we show our main result in Theorem 17, where the four conditions are not necessary but only sufficient. Nonetheless, they provide a guideline for the design of both linear and nonlinear ML models. As an example, the linear case is treated from this viewpoint in Corollary 18.

Theorem 14 *(Bolzano-Weierstrass) Every bounded sequence in $\mathbb{R}^k$ has a convergent subsequence.*

Proof See (Rudin, 1976, page 51). □

Theorem 15 *(Heine) A function $\boldsymbol{f} : \mathbb{R}^m \to \mathbb{R}^n$ satisfies $\lim_{\boldsymbol{x}\to\boldsymbol{p}} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{q}$ if and only if $\lim_{n\to\infty} \boldsymbol{f}(\boldsymbol{p}_n) = \boldsymbol{q}$ for every sequence $\{\boldsymbol{p}_n\}$ in $\mathbb{R}^m$ with $\boldsymbol{p}_n \neq \boldsymbol{p}$ and $\lim_{n\to\infty} \boldsymbol{p}_n = \boldsymbol{p}$.*

Proof See (Rudin, 1976, page 84). □

Lemma 16 *Let $\{\boldsymbol{x}_n\}$ be a bounded sequence in $(\mathbb{R}^d, \|\cdot\|)$. If each convergent subsequence of $\{\boldsymbol{x}_n\}$ converges to $\boldsymbol{x}$, then we have $\lim_{n\to\infty} \boldsymbol{x}_n = \boldsymbol{x}$.*

Proof Suppose that $\{\boldsymbol{x}_n\}$ does not converge to $\boldsymbol{x}$, i.e.,

$$\exists \epsilon > 0, \text{s.t.} \forall N > 0, \exists n > N, \text{s.t.} \|\boldsymbol{x}_n - \boldsymbol{x}\| > \epsilon.$$

Then there exists a subsequence $\{\boldsymbol{x}_{n_k}\}$ such that, for any $k > 0$, we have $\|\boldsymbol{x}_{n_k} - \boldsymbol{x}\| > \epsilon$. Since $\{\boldsymbol{x}_{n_k}\}$ is bounded, the Bolzano-Weierstrass theorem implies that it has a convergent subsequence $\{\boldsymbol{x}_{m_k}\}$. By the condition that each convergent subsequence of $\{\boldsymbol{x}_n\}$ has limit $\boldsymbol{x}$, we know that $\{\boldsymbol{x}_{m_k}\}$ converges to $\boldsymbol{x}$. But this contradicts $\|\boldsymbol{x}_{n_k} - \boldsymbol{x}\| > \epsilon$. □

As an answer to the question posed in the beginning paragraph of this subsection, the conditions (GLR-1,2,3,4) below constitute a guideline for designing regularization methods that satisfy the consistency principle.

Theorem 17 *The regularized model in Definition 2 is consistent with the original one if the minimization is formulated so that $\mathcal{L}(\cdot)$ and $\Omega(\cdot)$ satisfy*

(GLR-1) $\forall \boldsymbol{\beta} \in \mathbb{R}^d$, $\Omega(\boldsymbol{\beta}) \geq 0$;
(GLR-2) $\mathcal{L} : \mathbb{R}^d \to [0, +\infty)$ *is continuous;*
(GLR-3) $\hat{\boldsymbol{\beta}}_0$ *is the unique solution of (39a);*
(GLR-4) there exist constants $\epsilon > 0$ *and* $M > 0$ *such that* $\|\boldsymbol{\beta}\| > M$ *implies* $\mathcal{L}(\boldsymbol{\beta}) > \mathcal{L}(\hat{\boldsymbol{\beta}}_0) + \epsilon$.

Proof By Heine's theorem, it suffices to prove that for any positive sequence $\{\alpha_n\}$ satisfying $\lim_{n\to\infty} \alpha_n = 0$, we always have $\lim_{n\to\infty} \hat{\boldsymbol{\beta}}_{\alpha_n} = \hat{\boldsymbol{\beta}}_0$. We first show that $\{\hat{\boldsymbol{\beta}}_{\alpha_n}\}$ is bounded. For any $\alpha > 0$, (39) and (GLR-1) yield

$$\mathcal{L}(\hat{\boldsymbol{\beta}}_0) \leq \mathcal{L}(\hat{\boldsymbol{\beta}}_\alpha) \leq \mathcal{L}(\hat{\boldsymbol{\beta}}_\alpha) + \alpha\Omega(\hat{\boldsymbol{\beta}}_\alpha) \leq \mathcal{L}(\hat{\boldsymbol{\beta}}_0) + \alpha\Omega(\hat{\boldsymbol{\beta}}_0). \tag{42}$$

The squeeze theorem [2010, page 66] implies

$$\lim_{n\to\infty} \mathcal{L}(\hat{\boldsymbol{\beta}}_{\alpha_n}) = \mathcal{L}(\hat{\boldsymbol{\beta}}_0). \tag{43}$$

Suppose that $\{\hat{\boldsymbol{\beta}}_{\alpha_n}\}$ is unbounded, then for M in (GLR-4), we have infinite $\|\hat{\boldsymbol{\beta}}_\alpha\| > M$, and thus $\mathcal{L}(\hat{\boldsymbol{\beta}}_\alpha) > \mathcal{L}(\hat{\boldsymbol{\beta}}_0) + \epsilon$. This contradicts (43), hence (GLR-4) yields the boundness of $\{\hat{\boldsymbol{\beta}}_{\alpha_n}\}$. The Bolzano-Weierstrass theorem dictates that $\{\hat{\boldsymbol{\beta}}_{\alpha_n}\}$ has a convergent subsequence. Denote by $\boldsymbol{\theta}$ the limit of such a subsequence. (43) and (GLR-2) yield $\mathcal{L}(\boldsymbol{\theta}) = \mathcal{L}(\hat{\boldsymbol{\beta}}_0)$. Then the uniqueness of $\hat{\boldsymbol{\beta}}_0$ in (GLR-3) yields $\boldsymbol{\theta} = \hat{\boldsymbol{\beta}}_0$. Finally, the proof is completed by Lemma 16. □

By the counterexample (41), (GLR-4) is not a consequence of (GLR-3). Also, the regularized loss function $\mathcal{L}(\boldsymbol{\beta}) + \alpha\Omega(\boldsymbol{\beta})$ does not need to have a unique minimum point. By Theorem 17, each minimum point would converge to $\hat{\boldsymbol{\beta}}_0$ in the case of α approaching zero.

Theorem 17 provides a general way to the consistency of Tikhonov regularization for linear systems in Theorem 7.

Corollary 18 *Tikhonov regularization for linear systems is consistent, i.e.,*

$$\lim_{\alpha\to 0^+} \boldsymbol{x}_\alpha = \boldsymbol{x}^\oplus, \tag{44}$$

where $\mathrm{x}^\oplus$ *and* x_α *are the same as those in (10) and (23), and A has linearly independent columns.*

Proof The Tikhonov regularization has

$$\mathcal{L}(\boldsymbol{x}) = \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2, \quad \Omega(\boldsymbol{x}) = \|\boldsymbol{x}\|_2^2.$$

It suffices to verify (GLR-1,2,3,4) in Theorem 17. (GLR-1) holds trivially. By triangle inequalities, we have

$$\begin{aligned} & |\mathcal{L}(\boldsymbol{x}_1) - \mathcal{L}(\boldsymbol{x}_2)| \\ \leq\ & \|A\boldsymbol{x}_1 - A\boldsymbol{x}_2\|_2(\|A\boldsymbol{x}_1 - \boldsymbol{b}\|_2 + \|A\boldsymbol{x}_2 - \boldsymbol{b}\|_2) \\ \leq\ & \|A\|_2\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2(\|A\boldsymbol{x}_1 - \boldsymbol{b}\|_2 + \|A\boldsymbol{x}_2 - \boldsymbol{b}\|_2), \end{aligned}$$

which implies (GLR-2). Due to the linear independence of the columns of A, the case of multiple solutions in Theorem 3 is precluded. Then Theorem 2 implies (GLR-3). Finally, $\lim_{\|\boldsymbol{x}\|\to\infty} \mathcal{L}(\boldsymbol{x}) = +\infty$ leads to (GLR-4). □

For simple linear systems, we can often derive an explicit expression for $\hat{\beta}_\alpha$ in Tikhonov regularization, which is usually not possible for nonlinear systems or for a complicated penalty term $\Omega(\cdot)$. However, it is much easier to check (GLR-1,2,3,4) on $\mathcal{L}(\cdot)$ and $\Omega(\cdot)$ so that the consistency of the designed regularized model can be verified. This is a major significance of Theorem 17.

To sum up, the concept of regularization concerns at least three aspects. First, an ill-posed problem is regularized to admit one or multiple solutions. Second, the regularization makes it possible to balance accuracy and stability, providing the main mechanism to prevent overfitting. Third, the regularized problem should be consistent with the original problem in that the former approaches the latter in the limit of the vanishing regularization parameter, c.f. Definition 2. It is highly nontrivial to design a regularized model that simultaneously satisfies all of these three properties.

4 Generalization via Regularization

Out of the four design questions posed in Section 2.2, (ERM-3) is undoubtedly the most important one because it concerns the main utility of ML models. It is probably also the most difficult since the generalization performance is judged by results on data unavailable at the training stage.

The MSE reduction under Tikhonov in Section 3 shows that shrinking along small singular directions reduces variance; in non-linear models we cannot track singular vectors, but we can pursue the same stability via Lipschitz surrogates, gradient penalties, and margin-based losses. The remainder of this section summarizes these tools and Section 5-7 surveys concrete regularizers.

4.1 Bounding the Generalization Gap

To study nonlinear models in a way similar to that of linear models in Section 3.5, one would have to start with the *generalization gap*, i.e., the difference between the empirical risk $\mathcal{L}_{\text{ERM}}$ in (2) and the expected risk $\mathcal{L}_{\text{exact}}$ in (3),

$$g_{\mathcal{L}} := \|\mathcal{L}_{\text{ERM}} - \mathcal{L}_{\text{exact}}\|, \tag{45}$$

which is the most general evaluation metric for loss-based regularization. Due to the elusive nature of $\mathcal{L}_{\text{exact}}$, it is impossible to derive an analytic expression of the generalization gap

for all ML models. Instead, $g_{\mathcal{L}}$ should be specialized for the particular ML task at hand. For example, one might assume a specific analytic form of $\mathcal{L}_{\text{ERM}}$ or approximately calculate $g_{\mathcal{L}}$ using the results of test and training losses. There exist numerous studies on how to bound $g_{\mathcal{L}}$ in various scenarios.

4.2 Lipschitz Continuity

As pointed out in (MLM-2), data in ML always contain errors; in other words, there are always noises in the input space. A necessary condition for good generalization performance in ML is the stability of the model under input perturbations. Lipschitz continuity is a common tool for verifying the stability and analyzing the generalization performance (Weng et al., 2018; Luxburg and Bousquet, 2004).

Definition 3 A predictor function $f : \mathcal{X} \to \mathcal{Y}$ associated with an input metric space $(\mathcal{X}, d_X)$ and an output metric space $(\mathcal{Y}, d_Y)$ is said to be *Lipschitz continuous* if

$$\exists L \geq 0, \text{ s.t. } \forall \mathrm{x}, \mathrm{x}' \in \mathcal{X}, \quad d_y(f(\mathrm{x}), f(\mathrm{x}')) \leq L d_{\mathcal{X}}(\mathrm{x}, \mathrm{x}'), \tag{46}$$

where L is called the *Lipschitz constant*.
The Lipschitz constant is an upper bound of the ratio of changes in the output space to those in the input space, and thus measures the sensitivity of the model to input perturbations. Luxburg and Bousquet (2004) deduced a bound on $g_{\mathcal{L}}$ that depends on a Lipschitz constant for the Lipschitz classifier. By limiting the input space to a bounded metric space with a finite diameter $\text{diam}(\mathcal{X}) := \sup_{x,y \in \mathcal{X}} d(x, y)$, they proved that, for each $\epsilon \in (0, 1)$, (47) holds with probability at least $1 - \epsilon$,

$$\mathcal{L}_{\text{exact}}(f) - \mathcal{L}_{\text{ERM}}(f) \leq 2\mathfrak{R}_N(\mathcal{F}) + \sqrt{\frac{8 \log(2/\epsilon)}{N}}, \tag{47}$$

where $\mathcal{F}$ denotes the hypothesis space spanned by the allowed functions and $\mathfrak{R}_N(\mathcal{F})$ the Rademacher complexity of $\mathcal{F}$. The proportionality of $\mathfrak{R}_N(\mathcal{F})$ to $\left(L/\beta_{\text{margin}}\right)^N \text{diam}(\mathcal{X})$ was also shown. Their results suggest that models with smaller Lipschitz constants have better generalization performance.

Since the Lipschitz constant L can be approximated by the gradient of f, the model can be stabilized by penalizing $\|\nabla f(\mathrm{x})\|$, improving its generalization performance. We will discuss these regularization techniques in Section 6.2.

4.3 Simpler Models are Better in Generalization

Many theoretical and empirical studies suggest that model simplifications tend to improve the generalization performance, see, e.g., that by Neyshabur et al. (2017).

But how simple is simple? The first part of the answer concerns complexity measures of the model, including empirical-based measures (such as robustness of training error to parameter perturbation (Keskar et al., 2019) and the Liang et al. (2019)) and theoretical measures (such as VC dimension (Cherkassky et al., 1999; Mohri et al., 2018), Rademacher

complexity (Bartlett and Mendelson, 2002), norm-based measures (Neyshabur et al., 2015), and margins-based measures (Neyshabur et al., 2017)). VC dimension and Rademacher complexity are centered on the number of parameters while others are independent of it.

Bounds on the generalization gap have been derived in terms of the aforementioned complexity measures. For example, Mohri et al. (2018) gave a bound of $g_{\mathcal{L}}$ in the worst case. Suppose the range of the loss function $\mathcal{L}$ is in [0, 1]. For each $\epsilon \in (0, 1)$, (48) holds (Mohri et al., 2018, Thm. 3.3) with probability at least $1 - \epsilon$:

$$\begin{aligned} \mathcal{L}_{\text{exact}}(f) - \mathcal{L}_{\text{ERM}}(f) &\leq \sup_{f' \in \mathcal{F}} \left\{ \mathcal{L}_{\text{exact}}(f') - \mathcal{L}_{\text{ERM}}(f') \right\} \\ &\leq 2\mathfrak{R}_N(\mathcal{L}_{\mathcal{F}}) + \sqrt{\frac{\ln(1/\epsilon)}{2N}}, \end{aligned} \tag{48}$$

where N is the number of training samples in (2), $\mathcal{F}$ the hypothesis space of the allowed functions, $\mathcal{L}_{\mathcal{F}}$ a family of loss functions related to $\mathcal{F}$, and $\mathfrak{R}_N(\mathcal{L}_{\mathcal{F}})$ the Rademacher complexity of $\mathcal{L}_{\mathcal{F}}$, which can be bounded by $\mathfrak{R}_N(\mathcal{F})$.

As another example, Neyshabur et al. (2017) derived a generalization bound of the expected sharpness. For each $\epsilon \in (0, 1)$, (49) holds with probability at least $1 - \epsilon$,

$$\begin{aligned} &\mathbb{E}_{\boldsymbol{\delta}}[\mathcal{L}_{\text{exact}}(f(\mathrm{x}, \mathrm{w} + \boldsymbol{\delta}))] - \mathcal{L}_{\text{ERM}}(f(\mathrm{x}, \mathrm{w})) \\ &\leq \mathbb{E}_{\boldsymbol{\delta}}[\mathcal{L}_{\text{ERM}}(f(\mathrm{x}, \mathrm{w} + \boldsymbol{\delta}))] - \mathcal{L}_{\text{ERM}}(f(\mathrm{x}, \mathrm{w})) \\ &\quad + 4\sqrt{\frac{1}{N}\left(KL((\mathrm{w} + \boldsymbol{\delta})||Q) + \ln\frac{2N}{\epsilon}\right)}, \end{aligned} \tag{49}$$

where $f(\cdot, \mathrm{w})$ is a predictor learned from training data, $\boldsymbol{\delta}$ a random variable, P a distribution over predictors with weights of the form $\mathrm{w} + \boldsymbol{\delta}$, Q a prior distribution over the predictors independent of the training data, $KL(\cdot||\cdot)$ is the Kullback-Leibler (KL) divergence and the first two terms on the RHS of the inequality denote the expected sharpness.

Unfortunately, the heuristic of simpler models implying better generalization cannot explain the excellent performance of some models that have more parameters than training samples. In addition, some studies suggest that increasing the complexity may lead to better model generalization (Zhang et al., 2021; Kawaguchi et al., 2017; Li and Sompolinsky, 2021). We speculate that over-parametrization only leads to good generalization performance for a specific ML task where the inductive bias can capture the key features of the task, which may reconcile the contradiction of the aforementioned results and the title of this subsection.

4.4 A Roadmap for the Rest of this Survey

Hereafter, we focus on regularization methods based on the heuristic of reducing model complexity.

The L_0 regularization is probably the most straightforward way to simplify the model as it penalizes the number of parameters by setting $\Omega_{\ell_0}(\mathrm{w}) = \|\mathrm{w}\|_0 = \sum_{i=1}^{n} |w_i|^0 = n$, which forces most of the weight parameters to be zero. However, the resulting optimization is NP-hard and we have to relax this rigid and discontinuous constraint for many tasks.

The need for model parameters to vary *continuously* towards zero gives rise to regularizers with L_1 or L_2 norms as penalty terms. The Tikhonov regularization discussed in Section 3.4 is a prominent example with L_2 norm, which is also known as weight decay in ML and as ridge regression in statistics. L_1 regularization forces some weight parameters to be 0, thus preserving parameters of some important features. L_2 regularization keeps all the weight parameters small.

Following directly from the spectral cutoff discussed in Section 3.3 are sparse regularization methods in Section 5.2 and low-rank regularization strategies in Section 5.3.

Apart from model parameters, we can also penalize the parameterized function space; see Section 6.1.

It is commonly believed that a flat local minimum implies good generalization performance, as flat minima usually correlate to simple models (Keskar et al., 2019; Hochreiter and Schmidhuber, 1997). Regularization methods based on this heuristic will be reviewed in Section 7.

Although one can combat overfitting and improve model generalization by reducing the model complexity, we must maintain its expressiveness in order for the model to be useful. This is another important balance one has to strike.

We note that the loss-based regularizers surveyed in the remainder of this paper are complementary to *implicit* or *architecture* regularizers such as dropout (Srivastava et al., 2014), batch normalization (Dauphin and Cubuk, 2021), and data augmentation (Zhang et al., 2018). While those techniques regularize through the network structure or training procedure, the loss-based methods discussed in Sections 5–7 act by explicitly modifying or augmenting the objective function. Both families can be combined in practice.

The rest of the paper is a review of loss-based regularizers for ML that are specializations, augmentations, and/or further developments of the key ideas in Sections 3 and 4. These methods are classified according to the subject of the penalty in Figure 2 and briefly characterized in Table 1. The loss functions and evaluation metrics commonly adopted in regression and classification tasks are shown in Table 2.

5 Penalizing Weights

Penalizing weights drives the learning to smaller weights while integrating in the learning previous knowledges such as sparsity, low-rank, and smoothing. Consequently, the trained model tends to be simpler and have better generalization.

In this section, we discuss regularization methods in which a weight-based regularization term is added to the empirical risk loss (ERL) function. The subsections 5.1, 5.2, and 5.3 concern the L_2 penalty regularization, sparse regularization, and low-rank regularization, respectively.

5.1 L_2 Penalty Regularization

If a regularizer is too weak, its improvement on model generalization will be very limited. On the other hand, a regularizer with its effect too strong impairs the expression ability of the model. Path regularization and weight scale sifting invariant (WEISSI) regularization are two methods that could be used to strike a good balance.

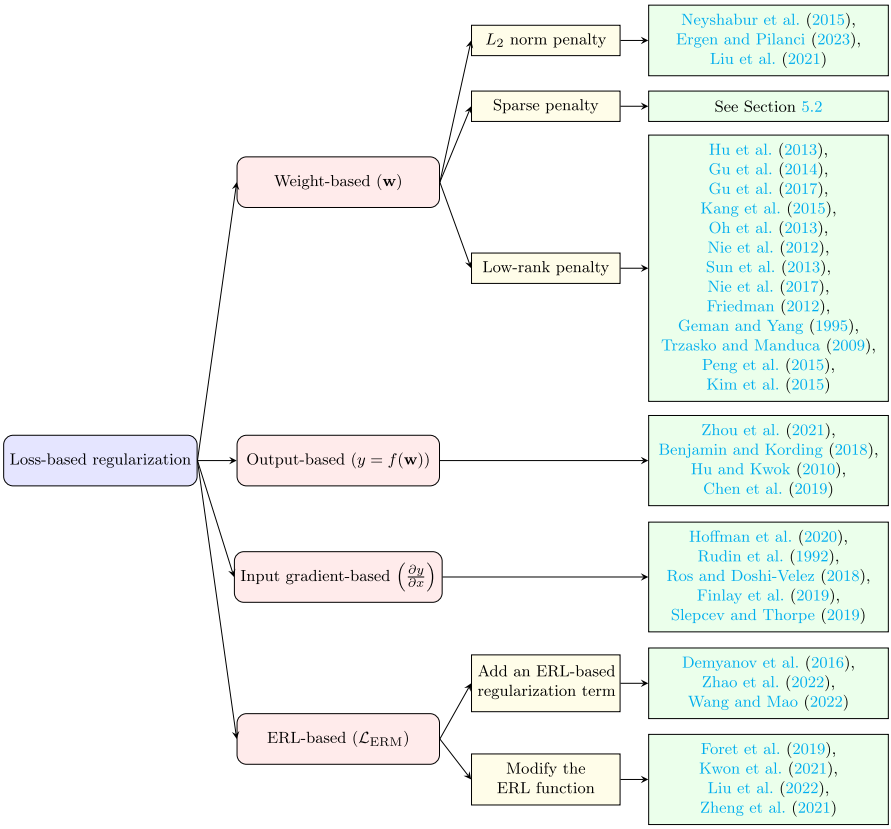


Fig. 2 Loss-based regularization methods

5.1.1 ℓ_2-Path Regularization

ℓ_2-path regularization (Neyshabur et al., 2015; Ergen and Pilanci, 2023) penalizes all weights in the network by

(1) multiplying the squared L_2 norm of all the parameters along each possible path from input to output;
(2) summing up the products over all paths;
(3) taking the squared root of the summation.

These steps yield

$$\Omega_{\ell_2\text{PR}}(\mathrm{w}) = \sqrt{\sum_{j_1,j_2,\cdots,j_L} \prod_{l=1}^{L} w^2_{l_{j_{l-1}j_l}}}, \tag{50}$$

where $l \in \{1, 2, \cdots, L\}$ denotes the lth layer of a L-layer network, $w_{l_{j_{l-1}j_l}}$ is the $j_{l-1}j_l$th entry of w_l. Unlike weight decay, ℓ_2-path regularization not only limits the size of weight,

Table 1 Loss-based regularization methods for model generalization

Penalty object	Taxonomy	Brief description
	L_2 norm penalty	It is based on L_2 regularization and realized by adding a regularization term that depends only on weights.
Weight	Sparse penalty	The idea is to constrain the weight vector to be sparse, realized by adding a sparse regularization term that depends only on weights.
	Low-rank penalty	The idea is to constrain the weight to be low-rank, realized by adding a low-rank regularization term that depends only on weights.
Output		Output-based penalty is to constrain the property of the output function, realized by adding a regularizationterm that depends on output functions.
Input gradient		The idea is to reduce a model's sensitivity to input perturbations, realized by adding a regularization term that depends on the gradient of output w.r.t. input.
ERL	Adding an ERL-based regularization term	It is based on reducing a model's sensitivity to input perturbations and/or the idea that a flat minimum improves the generalization performance, realized by adding a regularization term that depends on the ERL function.
	Modifying the ERL function	It is based on the idea that a flat minimum improves the generalization performance. It is realized by directly modifying the form of the ERL function itself.

Table 2 The loss functions and evaluation metrics commonly used in regression tasks and classification tasks

Tasks	Loss function	Evaluation metric
Regression	MSE, MAE, Huber loss, RSE, RAE, RMSE, MAPE	MSE, MAE, R Squared, Generalization gap
Classification	Cross-entropy loss, Hinge loss	Accuracy, Error Rate, (Average) Precision, F1 score, Top-k accuracy, Generalization gap

but also realizes a cross-layer coupling mechanism. As long as the product term in (50) is small enough, penalizing this product term will reduce some weights while allowing other weights to increase (Couellan, 2021). Consequently, the parameters are not only limited in size, but also have more degrees of freedom than when it is independently restricted at each layer, thus making the network more expressive and achieving the balance between accuracy and stability. In general, path regularization has the following form for $p \geq 1$:

$$\Omega_{\mathrm{PR}}(\mathrm{w}) = \left(\sum_{j_1, j_2, \cdots, j_L} \prod_{l=1}^{L} |w_{l j_{l-1} j_l}|^p \right)^{\frac{1}{p}}. \tag{51}$$

The ℓ_2-path regularization prevents the model from overfitting by penalizing all weights and is widely used in practice; see Figure 3 for an illustration of a three-layer network.

The optimization problem for (50) is non-convex. To resolve this problem, Ergen and Pilanci (2023) studied ℓ_2-path regularization on a deep parallel ReLU network architecture and transformed the optimization problem into a convex one. Moreover, they showed by experiments that ℓ_2-path regularization can improve the generalization without increasing the time complexity. When scanning a grid of α values, warm-starts let each solve begin from the previous iterate. With first-order or coordinate/proximal methods, the *per-iteration arithmetic* remains comparable; overall wall time then scales primarily with the number of grid points and early-stopping tolerance, rather than increasing the iteration cost itself.

5.1.2 WEISSI

According to Liu et al. (2021), weight decay is ineffective in deep neural networks where the activation function satisfies positive homogeneity, i.e., $\forall a > 0$, $\phi(ax) = a\phi(x)$. For such a network with $(L+1)$ layers, the model $\mathrm{y} = f(\mathrm{x}; \mathrm{W})$ can be expressed as

$$\forall l = 1, 2, \cdots, L, \quad \begin{cases} \mathrm{h}_l &= \phi(\mathrm{W}_l \mathrm{h}_{l-1} + \mathrm{b}_l); \\ \mathrm{y} &= \mathrm{W}_{L+1} \mathrm{h}_L, \end{cases} \tag{52}$$

where $\mathrm{h}_0 = \mathrm{x}$, the weight matrix $\mathrm{W}_l = [\mathrm{w}_{l,1}^T; \mathrm{w}_{l,2}^T; \cdots; \mathrm{w}_{l,s_l}^T]$, s_l is the size of the l-th layer. b_l is bias, and $\phi(\cdot)$ is the activation function. In the case of a network $\mathcal{N} = (\mathrm{W}_1, \mathrm{W}_2, \cdots, \mathrm{W}_{L+1})$ having a positively homogeneous activation function, a scalar can be extracted in each layer, i.e.,

$$\forall l = 1, 2, \cdots, L+1, \quad \phi(\mathrm{W}_l \mathrm{h}_{l-1} + \mathrm{b}_l) = a_l \phi(\overline{\mathrm{W}}_l \mathrm{h}_{l-1} + \overline{\mathrm{b}}_l),$$

where $\mathrm{W}_l = a_l \overline{\mathrm{W}}_l, \mathrm{b}_l = a_l \overline{\mathrm{b}}_l$. Then another network $\overline{\mathcal{N}} = (\overline{\mathrm{W}}_1, \overline{\mathrm{W}}_2, \cdots, \overline{\mathrm{W}}_{L+1})$ can be obtained so that

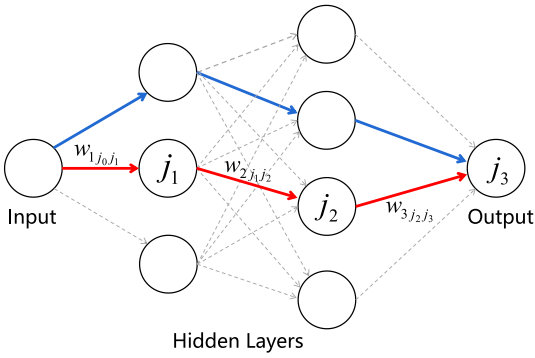


Fig. 3 Path regularization (Ergen and Pilanci, 2023). The network has a total of 12 paths, two of which are represented by the red and blue lines

$$\mathrm{y} = \left(\prod_{l=1}^{L+1} a_l\right) \cdot \overline{\mathrm{W}}_{L+1}\phi\left(\overline{\mathrm{W}}_L \mathrm{h}_{L-1} + \overline{\mathrm{b}}_L\right). \tag{53}$$

Provided that $\prod_{l=1}^{L+1} a_l = 1$ holds, $\overline{\mathcal{N}}$ is equivalent to $\mathcal{N}$, despite the seemlying different weight parameters. In other words, the weight scale between layers can be changed without changing the behavior of the network. Suppose weight decay were applied to these equivalent networks. The L_2 norm of weights of $\mathcal{N}$ would be smaller than that of $\overline{\mathcal{N}}$, which increases the likelihood of overfitting. To prevent this, Liu et al. (2021) set the scaling factor a_l to a normalization function $\eta(\mathrm{W}_l)$, transforming the model to

$$\mathrm{y} = \left(\prod_{l=1}^{L+1} \eta(\mathrm{W}_l)\right) \cdot \frac{\mathrm{W}_{L+1}}{\eta(\mathrm{W}_{L+1})}\phi\left(\frac{\mathrm{W}_L}{\eta(\mathrm{W}_L)}\mathrm{h}_{L-1} + \overline{\mathrm{b}}_L\right), \tag{54}$$

where $\eta(\mathrm{W}_l)$ is the $L_{p,q}$ norm of matrix W_l,

$$\|\mathrm{W}_l\|_{p,q} = \left(\sum_j \left(\sum_i |w_{l,ij}|^p\right)^{\frac{q}{p}}\right)^{\frac{1}{q}},$$

which leads to the following WEISSI term,

$$\Omega_{\mathrm{WEISSI}}(\mathrm{W}) = \lambda_e \prod_{l=1}^{L+1} \|\mathrm{W}_l\|_{2,2}^2 + \lambda_c \sum_{l=1}^{L+1} \left\| \frac{\mathrm{W}_l}{\|\mathrm{W}_l\|_{2,2}} \right\|_{1,1}, \tag{55}$$

where λ_e and λ_c are hyperparameters. The purpose of the first term is to impose a penalty on the total weight energy $\prod_{l=1}^{L+1} \eta(\mathrm{W}_l)$ of all layers and $\eta(\cdot)$ is derived from the $L_{2,2}$ norm. That of the second is to impose a penalty on the new weight $\frac{\mathrm{W}_L}{\eta(\mathrm{W}_L)}$, where the $L_{p,q}$ norm of the matrix $\frac{\mathrm{W}_L}{\eta(\mathrm{W}_L)}$ can be penalized by choosing $p = q = 1$.

Eq. (55) will not yield a small penalty even if the weight scale changes between layers. Therefore, this regularization can effectively prevent overfitting and improve generalization. In addition, experimental results by Liu et al. (2021) on standard image recognition datasets demonstrated that the generalization performance of WEISSI is better than weight decay in both standard and adversarial training.

5.2 Sparse Regularization

Sparse regularization terms added to the loss function tend to drive redundant weight parameters to zero and thereby remove unimportant connections or neurons. This reduction of the dimensionality of variable spaces simplifies the model, alleviates overfitting, and improves

generalization. They are widely used in high-dimensional data analysis such as mathematical statistics, machine learning, and bioinformatics.

As an example of sparse regularization, the L_0 regularization drives most of the elements of weights to zero, but transforms a convex optimization problem into a non-convex one. Since the non-convex optimization problem is an NP-hard problem (Natarajan, 1995), L_0 regularization is not practically efficient. L_1 regularization (Tibshirani, 1996), also called *lasso regression*, is an optimal convex approximation to L_0 regularization. However, the sparsity of L_1 regularization might still be insufficient for some applications. It is also sensitive to outliers and may lead to severely biased estimation (Fan and Li, 2001). As remedies, a series of more efficient sparse regularizers have been proposed, including alternative sparse regularization with less biased or unbiased estimates, as well as combinations of different sparse regularization; see Figure 4. The following subsections concern several sparse regularization methods in detail.

Group sparsity takes into account not only the removal of unimportant connections, but also that of isolated neurons. All outgoing connections from a single neuron are collectively considered as a *weight vector group*, where all elements are simultaneously required to be zero or non-zero. In the former case, the neuron is removed.

In the ℓ_1/ℓ_q*regularization*, an $L_q (1 \leqslant q \leqslant \infty)$ norm is used for weight parameters within each group while an L_1 regularization is used for all groups. An ℓ_1/ℓ_q regularization (Liu and Zhang, 2008; Liu and Ye, 2010) term takes the form:

$$\Omega_{\ell_1/\ell_q}(\mathrm{W}_G) = \sum_{\mathrm{g}\in G} (|\mathrm{g}|)^{1/q'} \|\mathrm{g}\|_{\ell_q}, \tag{56}$$

where g denotes a weight vector group of a single neuron, |g| its dimension, G the set of all weight vector groups. and q' the conjugate indices of q satisfying $\frac{1}{q'} + \frac{1}{q} = 1$ (assuming

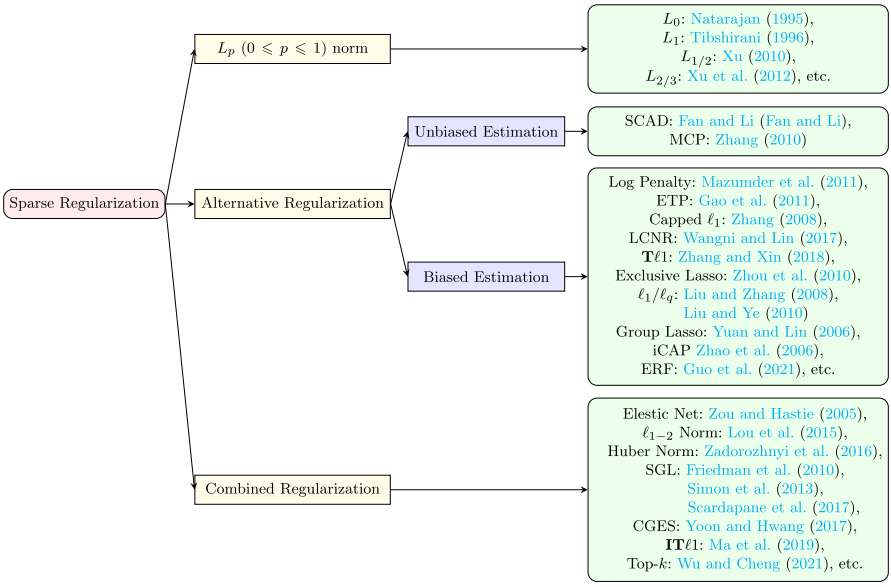


Fig. 4 Sparse regularization strategies

$\frac{1}{\infty} = 0$). The summation accounts for the L_1 regularization with $(|\mathrm{g}|)^{1/q'}$ being the weight of group g. For $q = \infty, 1, 2$, the ℓ_1/ℓ_q regularization reduces to the iCAP (Zhao et al., 2006), the standard L_1 regularization, and the *group lasso* (Yuan and Lin, 2006), respectively.

A group lasso term has the form

$$\Omega_{\ell_1/\ell_2}(\mathrm{w}) = \sum_{\mathrm{g}\in G} \sqrt{|\mathrm{g}|}\|\mathrm{g}\|_2.$$

By definition, group sparsity can achieve group-level sparsity. However, as shown in Figure 5(c), there is no sparsity within any remaining group, and this might not be optimal for some applications. Then we could combine group lasso with other sparse regularizations to form new schemes such as the sparse group lasso (SGL) (Friedman et al., 2010; Simon et al., 2013; Scardapane et al., 2017) and the integrated transformed $\ell 1(\mathrm{IT}\ell 1)$ (Ma et al., 2019). In these new schemes, not only certain neurons but also some connections of the remaining neurons are removed. In Figure 5 and Table 3, we compare lasso (Tibshirani, 1996), exclusive lasso (Zhou et al., 2010), group lasso (Yuan and Lin, 2006), SGL (Friedman et al., 2010; Simon et al., 2013; Scardapane et al., 2017), and CGES (Yoon and Hwang, 2017).

ERF regularization (Guo et al., 2021) is a method to improve the model sparsity by using an error function to approximate each term of the penalties, hence the name. An ERF regularization term takes the form $\Omega(\mathrm{w};\delta) = \sum \Omega(|w|_j;\delta)$, $\delta > 0$, where

$$\Omega(w;\delta) = \int_0^w e^{-\tau^2/\delta^2}\mathrm{d}\tau = \frac{\delta\sqrt{\pi}}{2}\mathrm{erf}\left(\frac{w}{\delta}\right). \tag{57}$$

and the standard *error function* is $\mathrm{erf}(w) = \frac{2}{\sqrt{\pi}}\int_0^w e^{-\tau^2}\mathrm{d}\tau$.

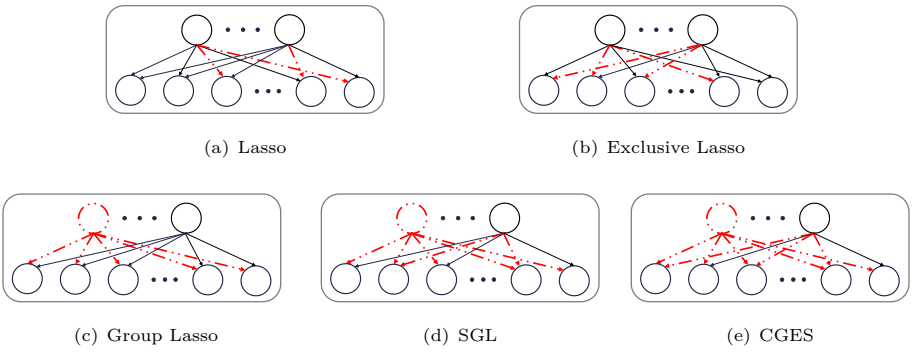


Fig. 5 A comparison of five sparse regularization strategies. The solid boxes show the connection diagrams of neurons in the $(l-1)$-th and l-th layer, and the red dashed lines denote the weights reduced to zero, i.e., the connections or neurons to be removed. **a Lasso.** It only removes a small number of neuron connections. **b Exclusive Lasso.** It removes a similar number of the output connections in the two neurons but does not remove neurons. In other words, it enforces similar sparsity within each group so that all groups have a similar number of non-sparse weight parameters. **c Group Lasso.** It removes all output connections of certain neurons so that this neuron can be eliminated. **d SGL.** It combines Group Lasso and Lasso by removing not only certain neurons but also some connections of the remaining neurons. **e CGES.** It combines Group Lasso and Exclusive Lasso by removing not only certain neurons but also a similar number of output connections of the remaining neurons

Table 3 A comparison of five sparse regularization strategies

Regularization	Location of sparsity	Remove connection	Remove a similar number of connections in different neurons	Remove neuron
Lasso	between connections	✓	✗	✗
Exclusive Lasso	within groups	✓	✓	✗
Group Lasso	between groups	✓	✗	✓
SGL	between groups and between connections	✓	✗	✓
CGES	between groups and within groups	✓	✓	✓

Theorem 19 *ERF regularization reduces to the L_1 regularization and the L_0 regularization in the limits of $\delta \to \infty$ and $\delta \to 0$, respectively.*

(1) $\lim_{\delta \to +\infty} \Omega(\mathrm{w}; \delta) = \|\mathrm{w}\|_1$;

(2) $\lim_{\delta \to 0^+} \frac{\Omega(\mathrm{w}; \delta)}{\delta} = \frac{\sqrt{\pi}}{2} \|\mathrm{w}\|_0$.

Proof Since $\Omega(\mathrm{w}; \delta)$ is the sum of the function $\Omega(|w|_j; \delta)$ of each component of w, it suffices to prove the limits for a scalar. When $w \neq 0$, let $t = \frac{w}{\delta}$, which goes to zero as $\delta \to +\infty$. Then we have

$$\lim_{\delta \to +\infty} \frac{\Omega(w; \delta)}{w} = \lim_{t \to 0} \frac{\int_0^t e^{-\tau^2} \mathrm{d}\tau}{t} = \lim_{t \to 0} e^{-t^2} = 1,$$

where the second step follows from the L'Hospital's rule. When $w = 0$, $\Omega(\frac{w}{\delta}; \delta) = 0 = w$. Therefore, $\Omega(\mathrm{w}; \delta) \to \|\mathrm{w}\|_1$ as $\delta \to +\infty$.

On the other hand, when $w \neq 0$, let $t = \frac{w}{\delta}$, which goes to $+\infty$ as $\delta \to 0^+$.

$$\lim_{\delta \to 0^+} \frac{\Omega(w; \delta)}{\delta} = \lim_{t \to +\infty} \int_0^t e^{-\tau^2} \mathrm{d}\tau = \frac{\sqrt{\pi}}{2}.$$

When $w = 0$, $\Omega(w; \delta) = 0$. Therefore, $\frac{\Omega(\mathrm{w}; \delta)}{\delta} \to \frac{\sqrt{\pi}}{2} \|\mathrm{w}\|_0$ as $\delta \to 0^+$. □

When δ is small, ERF regularization is relatively unbiased compared to other regularizers such as $L_{1/2}$ norm (Xu, 2010; Xu et al., 2012), log penalty (Mazumder et al., 2011), and ETP (Gao et al., 2011).

Top-$k$regularization is originally proposed for supervised feature selection in regression and classification tasks (Wu and Cheng, 2021). The regularizer consists of two parts: a given lead model and a sub-model with only k features. Specifically, the k most informative features are selected as a subset by $\mathrm{w}^{\max_k}$, the k largest weights, and inserted into the lead model. Thus the choice of the k features is made reasonably and the correlation between complex data is also captured. When lasso is used as the lead model, a sub-model with only

top-k features as input is plugged in. Then we have the regularization term (Wu and Cheng, 2021):

$$\Omega_{\text{top}-k}(\mathrm{w}) = \lambda_2 \mathcal{L}(f(\mathrm{w}^{\max_k}), D) + \lambda_1 \|\mathrm{w}\|_1, \tag{58}$$

where $\lambda_i \geq 0$ are the hyperparameters. The first term is the top-k regularization term, which is concerned with the predictive power of selected features, i.e., only the features with the highest weights are required to fit the data well. The second term is the lasso lead term.

Wu and Cheng (2021) also introduced top-k regularization methods with weight decay, elastic net, and deep neural networks as lead models. Top-k regularization introduces a sub-model into the prediction model, thereby improving the ability of balanced feature selection and data fitting. As such, they work well for supervised feature selection problems.

5.3 Low-Rank Regularization (LRR)

LRR is very popular for data processing analysis and has been used in a wide range of applications; see Table 6.

A low-rank matrix has a lot of redundant information, since the rank of a matrix indicates the correlation between its rows or columns. Low-rank regularization constrains the weight matrix in each layer to be low-rank, penalizing the redundant information in the matrix. Through low-rank constraints, the number of effective parameters is drastically reduced, thus preventing overfitting and improving the generalization of the model. Since a direct constraint on the rank of a matrix leads to an NP-hard optimization problem, it is necessary to replace the rank norm with some other norms. The regularization problem with the nuclear norm (see Table 6) is a convex approximation, but all singular values are penalized simultaneously. As such, there is a serious bias to matrices with small singular values since the largest singular values frequently quantify the important information. Many low-rank regularizers have been proposed to reduce the biased estimation of the rank norm, such as truncated nuclear norm (TNN) (Hu et al., 2013) and γ-nuclear norm (Kang et al., 2015). In addition, the sparse regularization methods such as SCAD (Fan and Li, 2001), MCP (Zhang, 2010), and ETP (Gao et al., 2011) mentioned in Section 5.2 can also be extended to low-rank regularization.

6 Penalizing Output and Input Gradient

The regularizers discussed in this section is based on the oberservation that it is the complexity of the predictor function that is ultimately regularized.

6.1 Output-based Penalty

Adding an output-based penalty term to the ERL function could be an effective regularization strategy. Manifold regularization is widely used in practice as an output-based technique that constrains properties of the learned function. We also mention several other strategies below.

6.1.1 $\mathbf{L_p}$ Norm

For training data with noisy labels, some common loss functions, such as the cross-entropy loss, will lead to model overfitting. To solve this problem, Zhou et al. (2021) introduced an output-based sparse regularization method that consists of two steps. The first step sharpens the output and the second imposes sparsity on the output. Specifically, the softmax function is first used to approximate the network output as a one-hot vector, and then a sparse $L_p(0 < p \leq 1)$ norm regularization is applied. The loss function is of the form

$$\sum L(f(\mathrm{x}_i), \tilde{y}_i) + \lambda \|f(\mathrm{x}_i)\|_p^p, \tag{59}$$

where $\lambda > 0$ is a hyperparameter, $f(\mathrm{x}_i)$ the output after sharpening in the first step, $L(f(\mathrm{x}_i), \tilde{y}_i)$ the symmetric loss function, and $\tilde{y}_i$ the noisy label of x_i corresponding to the label y_i. This loss function limits the output function to sparse and thus reduces the model complexity. Zhou et al. also proved that this method can fit data well and improve the robustness of any loss function by selecting an appropriate hyperparameter λ. Experimental results reported by Zhou et al. (2021) on CIFAR-10, CIFAR-100, and Clothing1M with ResNet and DenseNet architectures demonstrate that this method considerably enhances model generalization even in the presence of noisy labels and class imbalance.

6.1.2 Hilbert-Constrained Gradient Descent (HCGD)

Different from regularizers that exploit the natural gradient, HCGD (Benjamin and Kording, 2018) introduces a new learning rule that limits the change of the output function during the learning process and uses a more easily calculated L_2 norm to measure the incremental change of the network output function. An HCGD regularizer is of the form

$$\Omega_{\mathrm{HCGD}}(\mathrm{w}) = \|f_{\mathrm{w}_t} - f_{\mathrm{w}_t + \Delta \mathrm{w}}\|_2, \tag{60}$$

where Δw is the weight change and f_{w_t} the output function for the weight parameter w_t. This form of $\Omega_{\mathrm{HCGD}}(\mathrm{w})$ effectively limits the complexity of the output function.

6.1.3 Topological Regularization (TopoReg)

In contrast to most regularizers being structure-agnostic, TopoReg (Chen et al., 2019) penalizes topological complexity to enforce structural simplicity.

Consider the binary classification problem of locating, within a compact and simply connected feature space $\mathcal{X} \subset \mathbb{R}^D$, the level set at value zero of a smooth classifier function $f : \mathcal{X} \to \mathbb{R}$,

$$S_f = f^{-1}(0) = \{x \in \mathcal{X} | f(x) = 0\},$$

where the *classification boundary*S_f is a $(D-1)$-dimensional manifold that may consist of mulitple connected components. For each component c of S_f, there exists a unique pair of critical points (p_c, q_c) where p_c is a local minimum and q_c is a saddle point. Based on the theory of persistent homology (Edelsbrunner et al., 2002), this *persistence pairing*, i.e., the

association of (p_c, q_c) with $c \subset S_f$, can be shown to be injective. Consequently, the importance of a component c can be measured by its *robustness* (Chen et al., 2019),

$$\xi(c) = \min\{|f(p_c)|, |f(q_c)|\}. \tag{61}$$

Intuitively, $\xi(c)$ is the minimum amount of perturbation to f so that c stops being a connected component of S_f.

Chen et al. (2019) proposed to simplify the topological structure of S_f by penalizing on each connected component whose robustness is not the maximal, yielding

$$\Omega_{\text{TopoReg}}(\mathrm{w}) = L_{\mathcal{F}}(f(\cdot, \mathrm{w})) = \sum_{c \in \mathcal{C}(S_f)} \xi(c)^2, \tag{62}$$

where $\mathcal{C}(S_f)$ is the set of all connected components except the most robust of S_f. Results of numerical experiments show that TopoReg outperforms many existing methods. Finally, the idea of topological regularizer is general in that it also applies to higher dimensions and unsupervised contexts.

6.2 Input Gradient-based Penalty

The input gradient, i.e., the derivative of the output w.r.t. input, represents the extent to which the output changes due to input perturbations. To reduce model sensitivity to input perturbations, one can constrain the input gradient by adding a regularization term to the original ERL function.

As a typical class of regularization methods, input gradient regularization (Hoffman et al., 2020; Finlay et al., 2019) is of the form

$$\Omega_{\text{GR}}(\mathrm{w}) = \Omega\left[\nabla_{\mathrm{x}} f(\cdot, \mathrm{w})\right],$$

where different forms of Ω leads to different methods. For example, the regularizer $\Omega_{\text{GR}}(\mathrm{w}) = \|\nabla_{\mathrm{x}} f(\cdot, \mathrm{w})\|_p$ is called an L_p *regularizer*. For $p = 1, 2, \infty$, the L_p regularization is the total variation regularization (Rudin et al., 1992), the Tikhonov regularization (Ros and Doshi-Velez, 2018), and the Lipschitz regularization (Finlay et al., 2019), respectively. The Lipschitz regularization is an effective method that is often used to improve model robustness and generalization (Sokolić et al., 2017). A combination of L_p regularizations can also be used as regularizers (Finlay et al., 2019). Recent research on semi-supervised learning has shown that a higher p-norm of the gradient is needed to improve generalization (Slepcev and Thorpe, 2019). *Jacobian regularization* is the input gradient-based regularization with $\Omega_{\text{GR}}(\mathrm{w}) = \|\nabla_{\mathrm{x}} f(\cdot, \mathrm{w})\|_F^2$, where $\|\cdot\|_F^2$ denotes the Frobenius norm of a matrix. To alleviate the high cost of computating Jacobian matrices, an efficient algorithm is proposed in Hoffman et al. (2020) for solving the optimization problem with Jacobian regularization terms. In Hoffman et al. (2020), the Jacobian regularization is experimentally demonstrated to enhance the generalization on both visible and invisible domains. It can also be combined with other regularization strategies to improve model generalization.

7 Penalizing ERL Function

Adding an ERL-based penalty term to the original loss function or directly modifying the ERL function are both effective regularization techniques; see Figure 6. Demyanov et al. (2016) proposed the loss-invariant backpropagation (LIB) regularization to reduce the model's sensitivity to input perturbations by penalizing the gradient of the ERL function. Several other methods are based on the understanding that flat local minima of the ERL lead to better generalization. These methods are described in detail below.

7.1 Adding an ERL-based Regularization Term

7.1.1 Gradient Norm

Zhao et al. (2022) proposed a regularizer that penalizes the gradient norm of ERL w.r.t. the weights,

$$\Omega_{\mathrm{GN}}(\mathrm{w}) = \|\nabla_{\mathrm{w}}\mathcal{L}_{\mathrm{ERM}}(\mathrm{w})\|_p, \tag{63}$$

where $\|\cdot\|_p$ denotes the L_p norm; in practice, the L_2 norm is widely used. The optimization problem for (63) concerns the approximation of the Hessian matrix of $\mathcal{L}_{\mathrm{ERM}}(\mathrm{w})$ with a local Taylor expansion,

$$\begin{aligned} {}_{\mathrm{w}}\mathcal{L}_{\mathrm{Ob}}(\mathrm{w}) &= {}_{\mathrm{w}}\,\mathcal{L}_{\mathrm{ERM}}(\mathrm{w}) + {}_{\mathrm{w}}\,\Omega_{\mathrm{GN}}(\mathrm{w}) \\ &= (1-\frac{\lambda}{\mu}){}_{\mathrm{w}}\mathcal{L}_{\mathrm{ERM}}(\mathrm{w}) \\ &+ \frac{\lambda}{\mu}{}_{\mathrm{w}}\,\mathcal{L}_{\mathrm{ERM}}\left(\mathrm{w} + \mu\frac{{}_{\mathrm{w}}\mathcal{L}_{\mathrm{ERM}}(\mathrm{w})}{\|{}_{\mathrm{w}}\mathcal{L}_{\mathrm{ERM}}(\mathrm{w})\|}\right), \end{aligned}$$

where μ is small, $\lambda > 0$, and μ and λ are hyperparameters. The penalty imposed on the gradient norm of ERL functions leads to a small Lipschitz constant for the empirical risk, which further indicates a flat ERL landscape and thus better generalization. Indeed, Zhao et al. confirmed this by experiments.

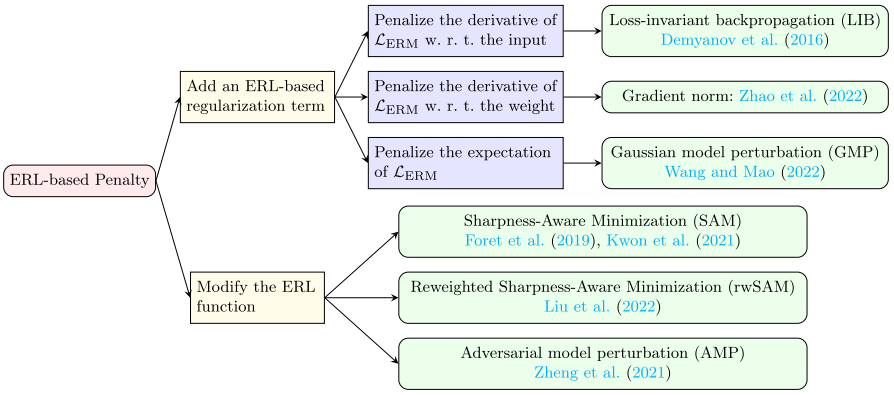


Fig. 6 ERL-based penalty methods

7.1.2 Gaussian Model Perturbation (GMP)

GMP is also based on the heuristic of a flat local minimum implying good model generalization. Wang and Mao (2022) related a generalized error bound to the flatness of the ERM solution by assuming Gaussian distribution $\mathcal{N}(0, \sigma^2 \mathrm{I}_d)$ of the perturbation $\boldsymbol{\Delta}$. GMP penalizes the expectation of the empirical risk change, yielding

$$\Omega_{\mathrm{GMP}}(\mathrm{w}) = \mathop{\mathbb{E}}_{\boldsymbol{\Delta} \sim \mathcal{N}(0, \sigma^2 \mathrm{I}_d)} \left[\mathcal{L}_{\mathrm{ERM}}(\mathrm{w} + \boldsymbol{\Delta}) - \mathcal{L}_{\mathrm{ERM}}(\mathrm{w}) \right],$$

where the expectation of $\mathcal{L}_{\mathrm{ERM}}(\mathrm{w} + \boldsymbol{\Delta})$ is approximated by k realizations of $\boldsymbol{\Delta}$:

$$\mathop{\mathbb{E}}_{\boldsymbol{\Delta} \sim \mathcal{N}(0, \sigma^2 \mathrm{I}_d)} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w} + \boldsymbol{\Delta}) = \frac{1}{k} \sum_{i=1}^{k} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w} + \boldsymbol{\delta}_i). \tag{64}$$

Wang and Mao (2022) not only explained theoretically the relationship between better generalization and flatter minima of loss functions, but also demonstrated by experiments that GMP has a better generalization performance than most regularizers.

7.2 Modifying the ERL Function

7.2.1 Sharpness-Aware Minimization (SAM)

To find the parameter that restricts the average minimum of the loss function within a certain neighborhood, Foret et al. (2019) proved an upper bound of generalization:

$$\mathcal{L}_{\mathrm{exact}}(\mathrm{w}) \leq \max_{\|\boldsymbol{\delta}\|_p \leq \beta} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w} + \boldsymbol{\delta}) + h\left(\frac{\|\mathrm{w}\|_2^2}{\beta^2} \right), \tag{65}$$

where $\beta \geq 0$ is a hyperparameter, $p \geq 1$, $\|\boldsymbol{\delta}\|_p \leq \beta$ is the maximum domain of parameter increments, and $h : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly increasing function. The RHS of (65) is

$$\left[\max_{\|\boldsymbol{\delta}\|_p \leq \beta} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w} + \boldsymbol{\delta}) - \mathcal{L}_{\mathrm{ERM}}(\mathrm{w}) \right] + \mathcal{L}_{\mathrm{ERM}}(\mathrm{w}) + h\left(\frac{\|\mathrm{w}\|_2^2}{\beta^2} \right),$$

where the terms within the square brackets is the sharpness of the ERL function and we get the L_2 regularization term with a hyperparameter $\lambda > 0$ if the last term is replaced by $\frac{\lambda}{2}\|\mathrm{w}\|_2^2$

.

SAM penalizes the sharpness of ERL, yielding the problem

$$\min_{\mathrm{w}} \max_{\|\boldsymbol{\delta}\|_p \leq \beta} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w} + \boldsymbol{\delta}) + \frac{\lambda}{2}\|\mathrm{w}\|_2^2. \tag{66}$$

Although SAM improves model generalization, the sharpness of loss function is unfortunately dependent on the weight scaling (Dinh et al., 2017). Assuming that any invertible scaling operator A in weight parameter space does not change the loss function; for two neural networks with weight parameters w and Aw, they have the same generalization gap but different sharpness.

To resolve this dependency issue of SAM, Kwon et al. (2021) proposed the adaptive sharpness-aware minimization (ASAM).

Definition 4 Denote by $\{T_{\mathrm{w}}|\mathrm{w}\in\mathbb{R}^k\}$ a family of invertible linear operators on $\mathbb{R}^k$. For a weight w, T_{w}^{-1} is a *normalization operator* of w if $T_{A\mathrm{w}}^{-1}A = T_{\mathrm{w}}^{-1}$ for any invertible scaling operator $A\in\mathbb{R}^{k\times k}$.

Then the adaptive sharpness can be defined as

$$\max_{\|T_{\mathrm{w}}^{-1}\boldsymbol{\delta}\|_p\le\beta} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w}+\boldsymbol{\delta}) - \mathcal{L}_{\mathrm{ERM}}(\mathrm{w}). \tag{67}$$

In (Kwon et al., 2021, Thm. 1), it is shown that any invertible scaling operator A in the weight parameter space does not change the loss function and the adaptive sharpness at w and Aw are the same. This conclusion leads to the ASAM optimization

$$\min_{\mathrm{w}} \max_{\|T_{\mathrm{w}}^{-1}\boldsymbol{\delta}\|_p\le\beta} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w}+\boldsymbol{\delta}) + \frac{\lambda}{2}\|\mathrm{w}\|_2^2. \tag{68}$$

The normalization operator ensures the independence of the adaptive sharpness of the ERL function from weight rescaling and adaptive sharpness has a stronger correlation with generalization gap than sharpness. As a result, ASAM is better than SAM in terms of generalization (Kwon et al., 2021).

7.2.2 Reweighted Sharpness-Aware Minimization (rwSAM)

Based on SAM, Liu et al. (2022) proposed the rwSAM for self-supervised learning. Consider a weight vector $\mathrm{a}\in\mathbb{R}^n$, and a weighted ERL function $\mathcal{L}_{\mathrm{RSAM}}(\mathrm{w}) = \frac{1}{n}\sum_{j=1}^{n} a_j l(x_j,\mathrm{w})$. The sharpness-aware minimization problem is formulated based on $\mathcal{L}_{\mathrm{RSAM}}(\mathrm{w})$. By adjusting the weight vector a, one gets a flatter loss landscape on a small amount of data; rwSAM generalizes well even for unbalanced data.

7.2.3 Adversarial Model Perturbation (AMP)

Zheng et al. (2021) proposed an alternative AMP loss without directly minimizing the ERL function. The AMP loss refers to the ERL in the worst case of adding norm-bounded perturbation to each point in the weight parameter space,

$$\mathcal{L}_{\mathrm{AMP}}(\mathrm{w}) = \max_{\boldsymbol{\delta}\in\mathrm{B}(0;\epsilon)} \mathcal{L}_{\mathrm{ERM}}(\mathrm{w}+\boldsymbol{\delta}), \tag{69}$$

where the hyperparameter ϵ has a small positive value, and $\mathrm{B}(0;\epsilon)$ denotes a sphere with center 0 and radius ϵ. Zheng et al. proved theoretically that AMP leads to a more flat local minimum and thus improves the generalization. According to (Zheng et al., 2021, Thm. 2), since the gradient of empirical risk w.r.t. weight parameters is penalized in AMP, the local minimum of the empirical risk can be found efficiently; furthermore, the local minimum features a small gradient norm, implying that the loss function is flat near the minimum.

8 A Summary in Tables

Many well-established regularizers have been widely used in various ML tasks. In this paper, however, we have focused on recent regularization methods.

In Table 4, we sum up the selection of a loss-based regularizer according to some practical criteria such as data regime, smoothness, convexity, budget, and interpretability. These guidelines complement the taxonomy by turning overlapping characteristics into concrete choices.

We summarize in three tables the regularizers discussed in Sections 5, 6, and 7.

• Table 5 concerns common sparse regularizers. It is known that non-convex regularizers yield more sparse solutions than convex regularizers.

• Table 6 concerns common low-rank regularizers. The nuclear norm is a classic method, but may lead to significant biases. Some non-convex low-rank regularizers can be used to alleviate this problem.

• Table 7 concerns regularization techniques based on penalizing the output, the input gradient, and the ERL function.

Finally, we test a number of regularization techniques based on the ERL function, including SAM, AMP, gradient norm, and GMP, with the ERM method as the baseline. Results of the classification accuracy of these methods on different CNN models are listed in Table 8, where those of SAM and the gradient norm method are taken from Zhao et al. (2022). For the VGG16 model (Simonyan and Zisserman, 2014), the performance of the ERM is improved by all regularization techniques, with AMP and GMP being the best. For the

Table 4 Selection guide for loss-based regularizers (illustrative, not exhaustive)

Family	Use when	Avoid when	Notes
L_2/ridge (Tikhonov)	Multicollinearity; need stability; dense signals	Strong sparsity desired	Cheap path via warm starts
L_1/elastic-net	True sparsity; feature selection needed	Features highly correlated and dense	Mix with L_2 if collinear
Gradient penalties	Robustness/ smoothness matter; labels clean	Targets non-smooth by design	Controls Lipschitz constants
Margin/label shaping	Class imbalance; calibration needed	Base loss interpretability paramount	Often improves calibration
Low-rank/trace	Compression/ latent structure	Full-rank signals crucial	Heavier optimization

Table 5 Sparse regularization techniques. In the table, VS stands for variable selection and IC stands for image classification

Regularization	Formula	Convex?	Main application tasks	Characteristic
Natarajan (1995) L_0	$\Omega_{\ell_0}(\mathrm{w}) = \|\mathrm{w}\|_0 = \sum_{i=1}^{n} \|w_i\|^0$	No	VS, Compressed Sensing	Optimization is NP-hard.
Tibshirani (1996) L_1	$\Omega_{\ell_1}(\mathrm{w}) = \|\mathrm{w}\|_1 = \sum_{i=1}^{n} \|w_i\|$	Yes	VS, Sparse Signal Reconstruction	Severely biased estimation
Xu (2010); Xu et al. (2012) L_p	$\Omega_{\ell_p}(\mathrm{w}) = \|\mathrm{w}\|_p = \left(\sum \|w_i\|^p\right)^{\frac{1}{p}}$	No	VS, Sparse Signal Reconstruction	Small biased estimation
Fan and Li (2001) SCAD	$\Omega(\mathbf{w};\lambda,\gamma) = \sum \begin{cases} \lambda\|w_i\|, & \|w_i\| \le \lambda, \\ \frac{2\gamma\lambda\|w_i\| - w_i^2 - \lambda^2}{2(\gamma-1)}, & 0 < \lambda < \|wi\| < \gamma\lambda, \gamma > 2 \\ (\gamma+1)\lambda^2/2, & \|w_i\| \ge \lambda \end{cases}$	No	VS	Completely unbiased estimation with two model parameters.
Zhang (2010) MCP	$\Omega(\mathrm{w};\lambda,\gamma) = \sum \begin{cases} \lambda\|w_i\| - w_i^2/(2\gamma) & \|w_i\| \le \gamma\lambda, \\ \gamma\lambda^2/2, & \|w_i\| > \gamma\lambda, \quad \lambda > 0, \gamma > 1 \end{cases}$	No	VS	Completely unbiased estimation with two model parameters.
Mazumder et al. (2011) Log penalty	$\Omega(\mathrm{w};\gamma) = \sum \Omega(w_i;\gamma) = \sum \frac{\log(\gamma\|w_i\| + 1)}{\log(\gamma+1)}, \gamma > 0$	No	VS, Sparse Signal Reconstruction	Small biased estimation with one model parameter.
Gao et al. (2011) ETP	$\Omega(\mathrm{w};\gamma) = \sum \Omega(w_i;\gamma) = \sum \frac{1-\exp(-\gamma\|w_i\|)}{1-\exp(-\gamma)}, \gamma > 0$	No	VS, Classification	Small biased estimation with one model parameter.
Zhang (2008) Capped ℓ_1	$\Omega(\mathrm{w};\gamma) = \sum \min(\|w_i\|, \gamma), \gamma > 0$	No	VS, Image Recovery	Only regularize weights below a certain threshold, hence has small biased estimation.
Wangni and Lin (2017) LCNR	$\Omega(\mathrm{w};\gamma) = \mu \sum \min_i(\|w_i\|, \gamma) + \nu \sum \max_i(\|w_i\|, \gamma), \gamma > 0$	No	3D Shape Recovery, Compressed Sensing	A small regularization factor for large weights, therefore with small biased estimation.
Zhang and Xin (2018) Tℓ1	$\mathrm{T}\ell1(\mathrm{w};\gamma) = \sum \Omega(w_i;\gamma) = \sum \frac{(\gamma+1)\|w_i\|}{\gamma + \|w_i\|}, \gamma > 0$	No	IC, Image Segmentation, Compressed Sensing, Sparse Signal Reconstruction	Relatively unbiased estimation with one model parameter.

Table 5 (continued)

Regularization	Formula	Convex?	Main application tasks	Characteristic
Guo et al. (2021) ERF	–	Yes	Compressed Sensing, Sparse Signal Reconstruction, Super-Resolution Tasks	Relatively unbiased estimation
Zou and Hastie (2005) Elestic Net	$\Omega(\mathrm{w};\gamma) = \gamma\Omega_{\ell_2}(\mathrm{w}) + (1-\gamma)\Omega_{\ell_1}(\mathrm{w}), 0 < \gamma < 1$	Yes	VS, IC, Sparse Signal Reconstruction	Combine L_1 and L_2 norm and more useful for data that has multiple features and some correlation between them.
Lou et al. (2015) ℓ_{1-2} norm	$\Omega_{\ell_{1-2}}(\mathrm{w}) = \|\mathrm{w}\|_1 - \|\mathrm{w}\|_2$	No	Image Separation Problems: Hyperspectral Imaging, DOAS, Super-resolution Separation	Combine L_1 and L_2 norm with smaller biased estimation.
Zadorozhnyi et al. (2016) Huber norm	$\Omega(\mathrm{w};\gamma,\lambda) = \sum \begin{cases} \lvert w_i\rvert - \frac{\lambda}{4\gamma}, & \lvert w_i\rvert \geq \frac{\lambda}{2\gamma}, \\ \frac{\gamma}{\lambda} w_i^2, & \lvert w_i\rvert < \frac{\lambda}{2\gamma}, \end{cases} \quad \lambda,\gamma > 0$	No	IC	Equivalent to combining L_1 and L_2 norm.
Zhou et al. (2010) Exclusive Lasso	$\Omega_{\mathrm{EL}}(\mathrm{w}) = \frac{1}{2}\sum_{\mathrm{g}\in G} \|\mathrm{g}\|_1^2$	Yes	VS	Enforce similar sparsity within each group.
Yuan and Lin (2006) Group Lasso	$\Omega_{\ell_1/\ell_2}(\mathrm{w}) = \sum_{\mathrm{g}\in G} \sqrt{\lvert\mathrm{g}\rvert}\|\mathrm{g}\|_2$	Yes	IC, VS	Remove connections and neurons.
Friedman et al. (2010); Simon et al. (2013); Scardapane et al. (2017) SGL	$\Omega_{\mathrm{SGL}}(\mathrm{w}) = \Omega_{\ell_1/\ell_2}(\mathrm{w}) + \Omega_{\ell_1}(\mathrm{w})$	Yes	IC, VS	Combine Group Lasso and Lasso.
Yoon and Hwang (2017) CGES	$\Omega_{\mathrm{CGES}}(\mathrm{w}) = \Omega_{\ell_1/\ell_2}(\mathrm{w}) + \Omega_{\mathrm{EL}}(\mathrm{w})$	Yes	IC, VS	Combine Group Lasso and Exclusive Lasso.
Ma et al. (2019) ITℓ1	$\Omega_{\mathrm{IT}\ell 1}(\mathrm{w}) = \Omega_{\ell_1/\ell_2}(\mathrm{w}) + \mathrm{T}\ell 1(\mathrm{w})$	No	IC	Combine Group Lasso and Tℓ1
Wu and Cheng (2021) Top-k	–	Yes	IC, VS, Semantic Analysis, Entity Recognition	Combine a leading model and a subset of k features.

Table 6 Low-rank regularization techniques

Regularization	Formula	Convex?	Main application tasks	Characteristic
Rank norm	$\Omega_{\mathrm{r}}(\mathrm{W}) = \|\sigma(\mathrm{W})\|_0 = \sum_{i=1}^{k} \|\sigma_i(\mathrm{W})\|^0$	No	Matrix Completion: Image, Video Completion; Robust PCA: Background Modeling	Optimization problem is an NP-hard problem.
Nuclear norm (NN)	$\Omega_{\mathrm{NN}}(\mathrm{W}) = \|\sigma(\mathrm{W})\|_* = \sum_{i=1}^{k} \sigma_i(\mathrm{W})$	Yes	Matrix Completion: Image, Video Completion and Restoration; Robust PCA: Face Analysis, 3D Reconstruction, Photometric Stereo, Motion Segmentation; Recommender System: Behavior Analysis	Seriously biased estimation for small singular value matrices.
Hu et al. (2013) TNN	$\Omega_{\mathrm{TNN}}(\mathrm{W}) = \sum_{i=r+1}^{k} \sigma_i(\mathrm{W})$	No	Matrix Completion: Image, Video Completion and Restoration, Visual Tracking; Robust PCA: HDRI, Salient Object Detection, Face Analysis	Truncate larger singular values and obtain relatively small biased estimation.
Oh et al. (2013) PSNN	$\Omega_{\mathrm{PSNN}}(\mathrm{W}) = \sum_{i=N+1}^{k} \sigma_i(\mathrm{W})$	No	Robust PCA: Background Modeling, HDRI, 3D Reconstruction and Image Alignment	Truncate partial singular values and obtain relatively small biased estimation.
Nie et al. (2012) S-p norm	$\Omega_{\mathrm{Sp}}(\mathrm{W}) = \sum_{i=1}^{k} \sigma_i^p(\mathrm{W}), 0 < p < 1$	No	Matrix Completion: Image, Video Completion; Recommender System: Behavior Analysis	Relatively small biased estimation and more suitable for solving practical problems than TNN and PSNN.
Sun et al. (2013) CNN	$\Omega_{\mathrm{CNN}}(\mathrm{W};\gamma) = \sum_{i=r+1}^{k} \min(\sigma_i(\mathrm{W}), \gamma), \gamma > 0$	No	Robust PCA: Background Modeling, Background Separation	Only regularize singular values that are below a certain threshold, therefore has small biased estimation.
Nie et al. (2017) Capped S-p	$\Omega_{\mathrm{CSp}}(\mathrm{W};\gamma) = \sum_{i=r+1}^{k} \min(\sigma_i^p(\mathrm{W}), \gamma), \gamma > 0$	No	Matrix Completion: Image, Video Completion; Robust PCA: Background Modeling; Recommender Systems: Behavior Analysis	Both CNN and S-p norm are used for singular values, therefore has small biased estimation.
Kang et al. (2015) γ-NN	$\Omega_{\gamma\text{-NN}}(\mathrm{W};\gamma) = \sum_{i=1}^{k} \frac{(1+\gamma)\sigma_i(\mathrm{W})}{\gamma + \sigma_i(\mathrm{W})}, \gamma > 0$	No	Robust PCA: Background Separation, Shadow Removal, Anomaly Detection	Relatively small biased estimation with only one model parameter.

Table 6 (continued)

Regularization	Formula	Convex?	Main application tasks	Characteristic
Friedman (2012) Logarithm	$\Omega_{\text{Log}}(\mathrm{W};\gamma) = \sum_{i=1}^{k} \frac{\log(\gamma\sigma_i(\mathrm{W})+1)}{\log(\gamma+1)}, \gamma > 0$	No	Subspace Clustering: Image Clustering, Video Image Motion Segmentation	Relatively small biased estimation with only one model parameter.
Geman and Yang (1995) Geman	$\Omega_{\text{Geman}}(\mathrm{W};\gamma) = \sum_{i=1}^{k} \frac{\sigma_i(\mathrm{W})}{\sigma_i(\mathrm{W})+\gamma}, \gamma > 0$	No	Matrix Completion: Image Recovery; Robust PCA: Image Reconstruction	Relatively small biased estimation with only one model parameter.
Trzasko and Manduca (2009) Laplace	$\Omega_{\text{Lap}}(\mathrm{W};\gamma) = \sum_{i=1}^{k} \left(1-\exp\left(-\frac{\sigma_i(\mathrm{W})}{\gamma}\right)\right), \gamma > 0$	No	Matrix Completion: Image Recovery; Robust PCA: Image Reconstruction	Relatively small biased estimation with only one model parameter.
Peng et al. (2015) Log NN	$\Omega_{\text{LNN}}(\mathrm{W}) = \sum_{i=1}^{k} \log(\sigma_i(\mathrm{W})+1)$	No	Matrix Completion: Image Recovery	Relatively small biased estimation without any model parameter.
Gu et al. (2014, 2017) WNN	$\Omega_{\text{WNN}}(\mathrm{W}) = \lVert\mathrm{W}\rVert_{w\star} = \sum_{i=1}^{k} \omega_i\sigma_i(\mathrm{W}), \omega_i \geq 0$	Yes	Matrix Completion: Image Completion; Robust PCA: Image Denoising, Background Modeling, Foreground Extraction	Weight each singular value and effectively reduce biased estimation.
Kim et al. (2015) ERSV	$\Omega_{\text{EN}}(\mathrm{W}) = \sum_{i=1}^{k} \left(\sigma_i(\mathrm{W}) + \gamma\sigma_i^2(\mathrm{W})\right), \gamma > 0$	Yes	Robust PCA: Shadow Removal, Nonrigid Motion Estimation, Photometric Stereo, Background Modeling	Combine L_1 and L_2 norm of singular values with small biased estimations.

Table 7 Regularization techniques based on output, input gradient, and ERL function penalty

	Reference	Regularization	Main application tasks	Learning type			
				Supervised	Semi-supervised	Self-supervised	Unsupervised
Output	Zhou et al. (2021)	L_p norm	IC1, IS, OD	✓			
	Benjamin and Kording (2018)	HCGD	IC1	✓			
	Hu and Kwok (2010)	Manifold	IA, IC1, SA, SS		✓		
	Chen et al. (2019)	TopoReg	IC1	✓			
Input gradient	Hoffman et al. (2020)	Jacobian	IC1, IG	✓			✓
	Rudin et al. (1992)	Total Variation	IC1	✓			
	Ros and Doshi-Velez (2018)	Tikhonov	IC1, IG	✓			
	Finlay et al. (2019)	Lipschitz	IC1, IC2, Image Inpainting	✓	✓		
	Slepcev and Thorpe (2019)	L_p norm	IC1, IC2	✓	✓		
ERL	Foret et al. (2019)	SAM	IC1, MT	✓			
	Kwon et al. (2021)	ASAM	IC1, MT	✓			
	Liu et al. (2022)	rwSAM	IC1, Instance Segmentation, MT, Depth Estimation, Graphic Annotation	✓		✓	
	Zheng et al. (2021)	AMP	IC1, IS, OD	✓			
	Demyanov et al. (2016)	LIB	IC1	✓			
	Zhao et al. (2022)	Gradient norm	IC1, IS, OD	✓			
	Wang and Mao (2022)	GMP	IC1, IS, OD	✓			

In the following, *IA* image annotation, *IC1* image classification, *IC2* image clustering, *IG* image generation, *IS* image segmentation, *MT* machine translation, *OD* object detection, *SS* semantic segmentation, *SA* sentiment analysis

Table 8 Classification accuracy acc.(%) of the VGG16 model and VGG16 model with batch normalization regularization on CIFAR-10 and CIFAR-100 datasets (Krizhevsky et al., 2009)

Basic data augmentation is used for all methods, with samples randomly flipped in horizontal, padded by four pixels, and then cropped randomly to size 32 x 32.

Bold values indicate the highest classification accuracy among all compared methods for specific models and datasets

Model	VGG16		VGG16-BN	
Dataset	CIFAR-10	CIFAR-100	CIFAR-10	CIFAR-100
ERM	92.93	71.14	94.19	74.49
SAM (Foret et al., 2019)	93.09	71.38	94.76	75.77
AMP (Zheng et al., 2021)	**93.48**	72.66	94.72	75.32
Gradient norm (Zhao et al., 2022)	93.28	71.52	**95.12**	**75.96**
GMP (Wang and Mao, 2022)	93.45	**72.72**	94.83	75.89

VGG16 model with batch normalization, the gradient norm method outperforms all other methods.

The comparison in Table 8 is illustrative and not intended as a comprehensive benchmark. We encourage readers to select methods using the decision criteria summarized in Table 4 rather than generalizing from this small example.

9 Concluding Remarks

We have considered ML through the ERM perspective and discussed key regularization concepts in the theoretical setting of inner-product spaces. We have analyzed the effectiveness of Tikhonov regularizations for linear regularizers and employed several popular heuristics for our exposition of nonlinear regularizers. In particular, we propose a consistency principle that could be used to guide the design of both linear and nonlinear regularizers.

We reviewed a wide range of recent loss-based regularization methods, suggested a comprehensive taxonomy of them, and summarized their characteristics and main applications in several tables. We also tested a number of representative loss-based regularizers by standard benchmark tasks, demonstrating satisfactory performance on preventing overfitting and improving generalization.

Our review of current regularization literature leads to observations as follows.

(1) No loss-based regularizers work best for all tasks. Instead, one should carefully select loss functions and evaluation metrics according to the particularities of the task at hand.
(2) Most regularizers are designed for specific tasks and there is a lack of generic regularization methods that simultaneously work well for different tasks.
(3) There is a lack of regularizers that work well for more complex data sets such as cross-model data sets and/or streaming data sets.

Accordingly, we believe that it is an important and promising research topic to develop regularization methods that work well for multiple tasks and complex data sets.

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Declarations

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